

#1 · Safe Work Practices

What is Routine Practices in a lab setting?

Answer

Routine Practices means treating every patient's blood and body fluids as if they could carry an infection, no matter who the patient is. This means every specimen gets the same precautions every time. It removes the need to guess which samples are risky.

#2 · Safe Work Practices

List the standard PPE for handling blood specimens.

Answer

Standard PPE includes gloves, a lab coat or gown, and eye protection such as a face shield or safety glasses. A mask may be added if splashing is likely. The right PPE depends on the task and the risk of exposure.

#3 · Safe Work Practices

When should you change your gloves?

Answer

Change gloves between patients, when they become torn or visibly soiled, and after touching a contaminated surface. Gloves should never be washed and reused. Fresh gloves protect both the patient and the worker from cross contamination.

#4 · Safe Work Practices

What does WHMIS stand for?

Answer

WHMIS stands for Workplace Hazardous Materials Information System. It is Canada's system for classifying and labeling hazardous products used at work. It gives workers information they need to handle chemicals safely.

#5 · Safe Work Practices

What is an SDS?

Answer

SDS stands for Safety Data Sheet. It is a document that explains a chemical's hazards, how to store it safely, and what to do if there is a spill or exposure. Every hazardous product in a lab must have one available to staff.

#6 · Safe Work Practices

What information does an SDS section on first aid measures cover?

Answer

This section tells you what to do right away if someone is exposed to the chemical, such as through skin contact, eye contact, or inhalation. It lists specific steps like flushing eyes with water or seeking medical attention. Following it quickly can reduce injury.

#7 · Safe Work Practices

What does the WHMIS flame hazard symbol mean?

Answer

The flame symbol warns that the product is flammable and can catch fire easily. This includes flammable liquids, gases, or solids. Workers should keep these products away from heat, sparks, and open flames.

#8 · Safe Work Practices

What does the WHMIS corrosion symbol mean?

Answer

The corrosion symbol shows a substance that can burn skin or eyes on contact, or that can eat through metal. It looks like liquid dripping onto a hand and a surface. Gloves and eye protection are essential when handling these products.

#9 · Safe Work Practices

What does the biohazard symbol represent?

Answer

The biohazard symbol marks materials that could contain infectious agents, such as blood, body fluids, or used sharps. Anything with this label needs special handling and disposal. It warns workers to use precautions before touching the item.

#10 · Safe Work Practices

What does the WHMIS exclamation mark symbol mean?

Answer

The exclamation mark covers less severe hazards, such as skin or eye irritation, drowsiness, or allergic reactions. It is a caution that the product can still cause harm even though it is not as dangerous as corrosives or flame hazards. Workers should still read the label before use.

#11 · Safe Work Practices

How should a sharps container be filled and disposed of?

Answer

A sharps container should never be filled past the fill line, usually marked at about three quarters full. Once full, it is sealed and sent for proper biohazard disposal, never emptied or reused. Overfilling raises the risk of a needlestick injury.

#12 · Safe Work Practices

What should you do immediately after using a needle or lancet?

Answer

Dispose of the sharp immediately into a puncture proof sharps container located near the point of use. Never recap a used needle by hand. Immediate disposal is the best way to prevent an accidental needlestick.

#13 · Safe Work Practices

What is the difference between disinfection and sterilization?

Answer

Disinfection kills most germs on a surface but does not always kill bacterial spores. Sterilization destroys all forms of microbial life, including spores. Lab surfaces are usually disinfected, while some instruments require full sterilization.

#14 · Safe Work Practices

What concentration of bleach solution is commonly used to disinfect lab surfaces?

Answer

A common working solution is a 1 in 10 dilution of household bleach with water, giving about 0.5 percent sodium hypochlorite. This solution should be made fresh regularly since it loses strength over time. It is used to wipe down counters and equipment after spills or at the end of a shift.

#15 · Safe Work Practices

What is autoclaving used for?

Answer

Autoclaving uses pressurized steam at high heat to sterilize equipment and destroy all microorganisms including spores. It is often used for reusable instruments and certain waste before disposal. This is a true sterilization method, stronger than routine disinfecting.

#16 · Safe Work Practices

What is ergonomics and why does it matter in a lab?

Answer

Ergonomics is designing tasks and workstations to fit the worker's body, reducing strain and injury. In a lab, this means adjusting chair height, monitor position, and phlebotomy chair setup to avoid repetitive strain injuries. Good ergonomics protects workers from long term back, neck, and wrist problems.

#17 · Safe Work Practices

Name two common ergonomic risks for a lab assistant who draws blood all day.

Answer

Repeated bending and twisting at the waist can strain the lower back, and repeated wrist motion during venipuncture can lead to strain injuries like tendonitis. Awkward reaching for supplies is another common risk. Adjusting the workstation and taking breaks helps reduce these risks.

#18 · Safe Work Practices

What is the first step when you discover a small biological spill?

Answer

Alert coworkers nearby, put on appropriate PPE, and contain the spill so it does not spread further. Only after these steps do you begin cleaning it up according to the facility's spill protocol. Acting quickly and calmly limits exposure risk.

#19 · Safe Work Practices

What does a spill kit typically contain?

Answer

A spill kit usually has absorbent material, disinfectant, gloves, a scoop or tongs, and a bag for contaminated waste. Some kits also include eye protection and a warning sign. Having it ready means staff can respond to a spill without delay.

#20 · Safe Work Practices

What does the acronym RACE stand for in fire response?

Answer

RACE stands for Rescue, Alarm, Contain, and Extinguish or Evacuate. Rescue means moving anyone in immediate danger. Alarm means alerting others and activating the fire alarm, then containing the fire by closing doors before extinguishing it or evacuating.

#21 · Safe Work Practices

What does the acronym PASS stand for when using a fire extinguisher?

Answer

PASS stands for Pull, Aim, Squeeze, and Sweep. Pull the safety pin, aim the nozzle at the base of the fire, squeeze the handle, and sweep side to side. This method helps put out small fires safely and effectively.

#22 · Safe Work Practices

When should an incident report be filed?

Answer

An incident report should be filed any time there is a needlestick injury, chemical exposure, spill, patient identification error, or near miss. Even events that did not cause harm should be reported so patterns can be spotted. Reporting helps the facility fix unsafe conditions before someone gets hurt.

#23 · Safe Work Practices

What should you do right after a needlestick injury?

Answer

Immediately wash the area with soap and water, then report the injury to your supervisor right away. Follow the facility's exposure protocol, which usually includes testing and possible treatment. Quick action lowers the chance of infection and starts the required follow up care.

#24 · Safe Work Practices

What is a worker's right to refuse unsafe work?

Answer

Workers in Canada have the legal right to refuse work they believe is unsafe to themselves or others. They must report the concern to a supervisor, who investigates before the work continues. This right protects staff from being forced into dangerous situations.

#25 · Safe Work Practices

What are the three main parts of the internal responsibility system in workplace safety?

Answer

The internal responsibility system means workers, supervisors, and employers all share responsibility for safety. Workers must follow procedures and report hazards, supervisors must enforce safe practices, and employers must provide training and equipment. Safety works best when everyone plays their part.

#26 · Safe Work Practices

Why should hand hygiene be done even when gloves are worn?

Answer

Gloves can develop tiny tears that are not visible, and hands can become contaminated while removing gloves. Hand hygiene should happen before putting gloves on and right after taking them off. This double layer of protection lowers the risk of spreading germs.

#27 · Safe Work Practices

What is the correct order for removing PPE?

Answer

The general order is gloves first, then eye protection, then gown, and mask last. This order keeps the most contaminated items, the gloves, from touching your clean skin or clothing. Hands should be washed immediately after all PPE is removed.

#28 · Safe Work Practices

What does the term biohazardous waste include?

Answer

Biohazardous waste includes used sharps, blood soaked materials, cultures, and any specimen containers that held body fluids. This waste must go into labeled biohazard bags or sharps containers, never regular garbage. Special handling prevents exposure to workers and the public.

#29 · Safe Work Practices

What should you check on a chemical label before using a product?

Answer

Check the hazard symbols, the signal word such as danger or warning, and any handling precautions listed. Also confirm the product has not expired and matches what you expect to be using. This quick check can prevent using the wrong or unsafe chemical.

#30 · Safe Work Practices

What is the purpose of an eyewash station in a lab?

Answer

An eyewash station flushes chemicals or biological material out of the eyes quickly after an accidental splash. It should be used for at least fifteen minutes while holding the eyelids open. Workers should know its exact location before an emergency happens.

#31 · Safe Work Practices

What does the exploding circle WHMIS symbol indicate?

Answer

This symbol warns of a product that can explode if exposed to heat, shock, or friction. Gases under pressure or unstable compounds may carry this symbol. Extra caution is needed in storage and handling to avoid triggering a reaction.

#32 · Safe Work Practices

What does the skull and crossbones symbol mean on WHMIS labels?

Answer

This symbol marks a product that is acutely toxic and can cause serious harm or death even in small amounts. It may apply to poisons that harm through swallowing, skin contact, or inhalation. Strict precautions and limited exposure time are required.

#33 · Safe Work Practices

Why is proper labeling of secondary containers important?

Answer

When a chemical is transferred from its original container into a smaller bottle for daily use, that new container needs its own label. The label should show the product name and hazard information. Without it, the next person using the container has no way to know what is inside or how dangerous it is.

#34 · Safe Work Practices

What should a lab worker do before starting a new task involving an unfamiliar chemical?

Answer

The worker should read the SDS for that chemical and confirm they understand the hazards and required PPE. They should also check the label for any special storage or handling instructions. Skipping this step risks unexpected exposure.

#35 · Safe Work Practices

What is meant by engineering controls in workplace safety?

Answer

Engineering controls are physical changes to equipment or the workspace that reduce a hazard, such as safety needles, fume hoods, or sharps containers built into the wall. They are considered more reliable than relying on PPE alone. They remove the hazard at its source instead of just blocking exposure.

#36 · Safe Work Practices

How does administrative control differ from engineering control?

Answer

Administrative controls are changes to work procedures or policies, like scheduling rotations to limit chemical exposure time or requiring specific training before a task. Engineering controls physically change equipment, while administrative controls change how people work. Both are used together with PPE for full protection.

#37 · Safe Work Practices

What should you do if a coworker is exposed to a splash of a corrosive chemical on their skin?

Answer

Immediately flush the affected area with large amounts of water for at least fifteen minutes. Remove any contaminated clothing while flushing if it is safe to do so. Then seek medical attention and file an incident report.

#38 · Safe Work Practices

Why must lab coats be removed before leaving the lab area?

Answer

Lab coats can carry contamination from chemicals or biological material picked up during work. Wearing them outside the lab, such as in a break room, risks spreading that contamination to clean areas. They should be stored and laundered separately from personal clothing.

#39 · Safe Work Practices

What is the purpose of a Material Safety Data Sheet binder or WHMIS station in a lab?

Answer

This is a central, easily accessible location where all SDS documents for chemicals used in the lab are kept. Workers can quickly find hazard and first aid information during an emergency. Keeping it updated and accessible is a legal requirement under WHMIS.

#40 · Safe Work Practices

What does proper hand hygiene technique include?

Answer

Proper technique includes wetting hands, applying soap, scrubbing all surfaces including between fingers and under nails for at least fifteen to twenty seconds, then rinsing and drying thoroughly. Alcohol based hand sanitizer can be used when hands are not visibly soiled. Visible soiling always requires soap and water instead.

#41 · Safe Work Practices

What is a near miss and why should it still be reported?

Answer

A near miss is an event that could have caused injury or harm but did not, such as almost dropping a specimen or nearly missing a sharps container. Reporting it helps identify hazards before an actual injury occurs. It gives the facility a chance to fix the problem early.

#42 · Safe Work Practices

How should broken glass contaminated with blood be cleaned up?

Answer

Never pick up broken glass with bare hands. Use a dustpan, brush, or tongs while wearing gloves, and dispose of the glass in a designated sharps or broken glass container. The area should then be disinfected according to spill protocol.

#43 · Safe Work Practices

What does the term personal protective equipment donning mean?

Answer

Donning means putting on PPE in the correct order before starting a task, typically gown, mask, eye protection, then gloves. Doing it in the right order ensures no gaps in coverage. It is the opposite step of doffing, which is removing PPE safely.

#44 · Safe Work Practices

Why should you never eat or drink in a lab work area?

Answer

Food and drink can become contaminated by chemicals, blood, or other biological material present in the lab. Eating or drinking also increases the chance of accidentally touching your mouth with contaminated hands. Food and drink should only be consumed in designated break areas.

#45 · Safe Work Practices

What should be done with expired or unused chemicals in the lab?

Answer

They should be disposed of according to the facility's hazardous waste protocol, never poured down a regular sink or thrown in normal trash. The SDS often lists specific disposal guidance for that chemical. Following this prevents environmental contamination and safety hazards.

#46 · Safe Work Practices

What is the role of a supervisor when a worker refuses unsafe work?

Answer

The supervisor must investigate the concern right away, often together with a health and safety representative. Work in that area should stop until the hazard is assessed and resolved. The worker cannot be punished for making a genuine safety refusal.

#47 · Safe Work Practices

What is meant by chain of infection in the context of routine practices?

Answer

The chain of infection describes the steps germs need to spread, including a source, a way to leave the source, a way to travel, and a way to enter a new host. Routine practices work by breaking one or more links in this chain, such as hand hygiene or PPE. Breaking any single link can stop the spread of infection.

#48 · Safe Work Practices

Why is proper posture important when working at a lab bench for long periods?

Answer

Poor posture during long periods of pipetting or sample sorting can strain the neck, shoulders, and lower back over time. Keeping the spine neutral and the workstation adjusted to the right height reduces this strain. Taking short breaks to stretch also helps prevent chronic injury.

#49 · Safe Work Practices

What does the term biological safety cabinet protect against?

Answer

A biological safety cabinet protects the worker, the sample, and the environment from exposure to infectious aerosols or particles during certain procedures. It uses filtered airflow to contain contaminants inside the cabinet. It is used when handling specimens that may release airborne hazards.

#50 · Safe Work Practices

What is the first thing you should do if you are unsure whether a task is safe to perform?

Answer

Stop and ask a supervisor or senior coworker before proceeding, rather than guessing. It is always safer to pause and confirm the correct procedure. Reporting uncertainty early prevents mistakes that could harm you or a patient.

#51 · Equipment Instruments and Reagents

What is Kohler illumination used for on a microscope?

Answer

It sets up the light path so the specimen gets even, bright lighting with no glare or shadow. The technique focuses the field diaphragm and centers the condenser before viewing slides. Proper Kohler illumination gives the sharpest image and the best contrast at every magnification.

#52 · Equipment Instruments and Reagents

Name the three main lenses in a compound microscope light path, from specimen to eye.

Answer

Light passes through the objective lens first, then travels up the body tube to the eyepiece (ocular) lens. The condenser lens sits below the stage and focuses light onto the specimen before it ever reaches the objective. Together these lenses magnify and focus the image the user sees.

#53 · Equipment Instruments and Reagents

Why must a centrifuge always be balanced before running?

Answer

An unbalanced rotor spins with uneven weight on each side, which causes vibration and can crack tubes or damage the bearings. Balancing means placing a tube of matching weight directly across from every sample tube. A severely unbalanced centrifuge can even walk itself off the counter or injure a person nearby.

#54 · Equipment Instruments and Reagents

What does RCF stand for in centrifugation, and why does it matter more than RPM?

Answer

RCF stands for relative centrifugal force, the actual force pulling particles outward during spinning. Two centrifuges can spin at the same RPM but produce different RCF values because rotor radius differs between machines. Lab procedures specify RCF so results stay consistent no matter which centrifuge is used.

#55 · Equipment Instruments and Reagents

What is the purpose of a biosafety cabinet (BSC)?

Answer

A biosafety cabinet protects the worker, the specimen, and the environment from exposure to infectious aerosols. It uses directional airflow and HEPA filtration to contain particles generated while handling samples. Class II cabinets are the most common type found in clinical labs.

#56 · Equipment Instruments and Reagents

What does HEPA stand for, and what does it filter out of a biosafety cabinet's air?

Answer

HEPA stands for high efficiency particulate air. A HEPA filter traps at least 99.97 percent of particles as small as 0.3 microns, including most bacteria and many aerosol droplets. This is what allows a biosafety cabinet to recirculate or exhaust air safely.

#57 · Equipment Instruments and Reagents

How often should a biosafety cabinet be certified, and by whom?

Answer

A biosafety cabinet needs certification at least once a year, and also any time it is moved or serviced. A qualified outside technician checks airflow velocity, HEPA filter integrity, and cabinet containment during this process. The certification sticker on the cabinet should always show a current date before use.

#58 · Equipment Instruments and Reagents

What is an autoclave used for in the lab?

Answer

An autoclave uses pressurized steam at high temperature to sterilize equipment, glassware, and biohazard waste. A common cycle runs at 121 degrees Celsius and 15 psi for about 15 to 20 minutes. Chemical or biological indicator strips are run with each load to confirm sterilization actually happened.

#59 · Equipment Instruments and Reagents

What is a biological indicator in autoclave testing?

Answer

A biological indicator is a small vial or strip containing heat resistant bacterial spores, usually *Geobacillus stearothermophilus*. After the autoclave cycle, the indicator is incubated to see if any spores survived. If no growth appears, the cycle reached a temperature and time sufficient to kill even the toughest organisms.

#60 · Equipment Instruments and Reagents

Why must pipettes be calibrated on a regular schedule?

Answer

Calibration checks that a pipette delivers the exact volume it is set to, since seals and pistons wear down with repeated use. An out of calibration pipette can throw off dilutions, reagent volumes, and test results without anyone noticing. Most labs calibrate pipettes every three to six months using a gravimetric water weighing method.

#61 · Equipment Instruments and Reagents

What is the gravimetric method of pipette calibration?

Answer

The pipette dispenses distilled water onto an analytical balance, and the weight is converted to volume using the known density of water at that temperature. This is repeated several times to check both accuracy and precision. Because water weighs almost exactly 1 gram per milliliter at room temperature, the math stays simple.

#62 · Equipment Instruments and Reagents

What is the difference between accuracy and precision when checking equipment performance?

Answer

Accuracy means the results are close to the true or expected value. Precision means repeated measurements are close to each other, whether or not they are correct. A pipette can be precise but inaccurate if it consistently delivers the wrong volume with very little variation between tries.

#63 · Equipment Instruments and Reagents

What routine maintenance keeps an analytical balance reading accurately?

Answer

The balance should sit on a stable, vibration free surface and be leveled using its built in bubble level. It should be calibrated regularly with certified reference weights. The weighing pan and chamber should be kept clean and free of dust, spills, or static charge that could affect the reading.

#64 · Equipment Instruments and Reagents

What does a pH meter measure, and what must be done before each use?

Answer

A pH meter measures how acidic or basic a solution is using a glass electrode sensitive to hydrogen ion concentration. Before use it must be calibrated with at least two standard buffer solutions, commonly pH 4 and pH 7 or pH 7 and pH 10. Skipping calibration can give a reading that looks normal but is actually wrong.

#65 · Equipment Instruments and Reagents

How should a pH electrode be stored between uses?

Answer

The electrode tip should be kept moist in a storage solution or buffer, never dry. Letting the glass bulb dry out damages the sensing membrane and shortens the electrode's life. Most manufacturers sell a specific storage solution made for this purpose.

#66 · Equipment Instruments and Reagents

What is a spectrophotometer used for in the clinical lab?

Answer

A spectrophotometer measures how much light of a specific wavelength passes through a liquid sample, which is related to the concentration of a substance in it. Many chemistry analyzers use this principle internally to measure things like glucose or bilirubin. The instrument reads absorbance, and a higher absorbance usually means a higher concentration.

#67 · Equipment Instruments and Reagents

What is a blank in spectrophotometry, and why is it needed?

Answer

A blank is a solution containing everything in the test except the substance being measured, such as just the reagent and diluent. The instrument is zeroed against the blank so its absorbance reading starts at zero. This removes background interference so the final reading reflects only the substance of interest.

#68 · Equipment Instruments and Reagents

How is a fluorometer different from a spectrophotometer?

Answer

A spectrophotometer measures light that passes straight through a sample, while a fluorometer measures light that a sample re-emits after being excited by a different wavelength. Fluorometers are often more sensitive and can detect very low concentrations of certain substances. Both instruments still rely on comparing sample readings to known standards.

#69 · Equipment Instruments and Reagents

What is point of care testing (POCT)?

Answer

POCT means running a test right where the patient is, such as at the bedside or in a clinic, instead of sending the sample to the main lab. Common examples include glucose meters, urine dipsticks, and rapid strep tests. Because results come back fast, POCT devices still need quality control checks just like main lab instruments.

#70 · Equipment Instruments and Reagents

What quality control is required for a glucose meter used for patient testing?

Answer

Glucose meters need liquid quality control run at set intervals, usually daily or per shift, using both a low and a high level control. The results must fall within the manufacturer's expected range before patient results can be reported. Meters also get periodically checked against a reference lab method for accuracy.

#71 · Equipment Instruments and Reagents

What does an ECG (electrocardiogram) record?

Answer

An ECG records the electrical activity of the heart as it beats, picked up through electrodes placed on the skin. The tracing shows the timing and shape of each heartbeat, which helps identify rhythm problems or damage to heart tissue. A standard ECG uses 12 leads to view the heart's electrical activity from different angles.

#72 · Equipment Instruments and Reagents

What is a Holter monitor, and how is it different from a standard ECG?

Answer

A Holter monitor is a small portable ECG device the patient wears continuously, usually for 24 to 48 hours. It records the heart's rhythm during normal daily activity instead of just a short snapshot in the clinic. This helps catch irregular heartbeats that only happen occasionally and might be missed on a regular ECG.

#73 · Equipment Instruments and Reagents

What is the correct way to apply ECG electrodes for a clean recording?

Answer

The skin should be cleaned and lightly abraded to remove oil and dead skin, which improves electrical contact. Electrodes are placed at specific anatomical landmarks on the limbs and chest according to the standard 12 lead pattern. Loose or poorly placed electrodes cause artifact, which is unwanted noise in the tracing.

#74 · Equipment Instruments and Reagents

What is an equipment maintenance log, and why must it be kept up to date?

Answer

An equipment maintenance log is a written or electronic record of every cleaning, calibration, function check, and repair done on an instrument. It proves the equipment was working correctly on the days it was used to test patients. Accreditation inspectors regularly review these logs, and gaps or missing entries can be flagged as a compliance problem.

#75 · Equipment Instruments and Reagents

What should be recorded every time a piece of equipment fails a function check?

Answer

The date, time, what failed, who found it, and what corrective action was taken should all be written down. The equipment should be taken out of service until the problem is fixed and it passes a check again. Any patient results run on the malfunctioning equipment may need to be reviewed or repeated.

#76 · Equipment Instruments and Reagents

What is preventative maintenance, and how is it different from a repair?

Answer

Preventative maintenance is scheduled work done on a set timetable to keep equipment running well before anything breaks, like cleaning, lubricating, or replacing filters. A repair happens after something has already failed or is performing incorrectly. Following the preventative maintenance schedule reduces the number of unexpected breakdowns and repairs.

#77 · Equipment Instruments and Reagents

What does the expiry date on a reagent bottle actually guarantee?

Answer

The expiry date is the last day the manufacturer guarantees the reagent will perform as expected when stored properly. Using an expired reagent can give inaccurate results even if the liquid looks completely normal. Expired reagents should be discarded and never used for patient testing, even in an emergency.

#78 · Equipment Instruments and Reagents

How should reagents that require refrigeration be stored and monitored?

Answer

They must be kept within the manufacturer's stated temperature range, usually 2 to 8 degrees Celsius for refrigerated reagents. The refrigerator temperature should be checked and logged at least once daily. If the fridge temperature drifts out of range, the reagents inside may need to be quarantined until their stability can be confirmed.

#79 · Equipment Instruments and Reagents

What are signs that a reagent has gone bad, even before its expiry date?

Answer

Signs include a change in color, cloudiness or visible particles, an unusual odor, or a broken seal on the container. Quality control results that suddenly shift outside the expected range can also point to a bad reagent lot. Any reagent showing these signs should not be used and should be reported to a supervisor.

#80 · Equipment Instruments and Reagents

What information must be on a reagent label once it is opened or reconstituted?

Answer

The label should show the date it was opened or reconstituted, the new expiry date if it changed, and the initials of the person who prepared it. This tracks how long the reagent has actually been in use, since opening can shorten its shelf life. Without this information, staff cannot tell if the reagent is still safe to use.

#81 · Equipment Instruments and Reagents

What is the correct way to dispose of expired or unused chemical reagents?

Answer

They must be disposed of according to the safety data sheet (SDS) and local hazardous waste regulations, never poured down a regular sink drain unless the SDS specifically allows it. Many reagents need to go into labeled chemical waste containers for pickup by a hazardous waste company. Mixing incompatible chemicals together in one waste container can create dangerous reactions.

#82 · Equipment Instruments and Reagents

What is a Safety Data Sheet (SDS) and why does every reagent need one on file?

Answer

An SDS is a standardized document that lists a chemical's hazards, safe handling instructions, storage requirements, and emergency first aid steps. Every hazardous reagent used in the lab must have an accessible SDS, either in a binder or an electronic system. Staff refer to it during a spill, exposure, or when unsure how to store or dispose of a chemical.

#83 · Equipment Instruments and Reagents

What does reagent lot number tracking accomplish?

Answer

Every container of reagent comes from a manufactured batch called a lot, and each lot gets its own number. Recording which lot number was used for each test allows the lab to trace a problem back to a specific batch if something goes wrong. This is important if a manufacturer later issues a recall for that lot.

#84 · Equipment Instruments and Reagents

Why should new reagent lots be verified before being put into full use?

Answer

A new lot may perform slightly differently from the previous one even though it is the same product name. Running quality control samples and comparing results against the old lot confirms the new lot gives consistent, reliable results. This step catches manufacturing problems before they affect patient testing.

#85 · Equipment Instruments and Reagents

What basic electrical safety rule applies to any lab instrument with a frayed cord or exposed wiring?

Answer

The equipment should be unplugged immediately and taken out of service until it is inspected or repaired by qualified maintenance staff. Continuing to use damaged electrical equipment risks shock or fire. A tag should be placed on the equipment clearly stating it is out of service.

#86 · Equipment Instruments and Reagents

What is the purpose of a ground fault circuit interrupter (GFCI) outlet in a lab?

Answer

A GFCI outlet monitors the electrical current flowing through it and shuts off power almost instantly if it detects current leaking somewhere it should not go, such as through water or a person. This greatly reduces the risk of electric shock, especially near sinks or wet areas. Labs often require GFCI outlets anywhere liquids are used near powered equipment.

#87 · Equipment Instruments and Reagents

In simple electrical terms, what does voltage measure compared to current?

Answer

Voltage is the electrical pressure or force that pushes electric charge through a circuit, measured in volts. Current is the actual flow rate of that charge, measured in amperes (amps). Higher voltage pushing through a fixed resistance results in higher current flow.

#88 · Equipment Instruments and Reagents

Why is total magnification calculated by multiplying the objective and eyepiece power?

Answer

The eyepiece further magnifies the image that the objective lens has already magnified, so the two effects multiply together rather than add. For example, a 10x eyepiece with a 40x objective gives 400x total magnification. Knowing the total magnification helps a technologist judge cell size and identify structures correctly.

#89 · Equipment Instruments and Reagents

What is the oil immersion objective used for, and why is oil needed?

Answer

The oil immersion objective, usually 100x, is used for viewing very small structures like bacteria or fine cell detail at high magnification. Immersion oil has a refractive index similar to glass, which prevents light from bending and scattering as it leaves the slide. Without the oil, the image at this magnification would be too dim and blurry to use.

#90 · Equipment Instruments and Reagents

How should a microscope's oil immersion lens be cleaned after use?

Answer

The lens should be wiped gently with lens paper, sometimes lightly dampened with a proper lens cleaning solution, right after use so the oil does not dry and harden on the glass. Regular tissues or paper towels should never be used because they can scratch the lens coating. Leftover dried oil can permanently damage image quality if not cleaned promptly.

#91 · Equipment Instruments and Reagents

What does the numerical aperture of an objective lens describe?

Answer

Numerical aperture describes the lens's ability to gather light and resolve fine detail, essentially how sharp and clear the image can be. A higher numerical aperture generally means better resolution at higher magnifications. This is why oil immersion objectives, which have high numerical apertures, need oil to work properly.

#92 · Equipment Instruments and Reagents

What routine cleaning should be done on a centrifuge to prevent contamination and rust?

Answer

The rotor, buckets, and interior chamber should be wiped down regularly with an appropriate disinfectant to remove spills or broken glass from tubes. Any cracked tube holders or corroded metal parts should be reported and replaced. Regular cleaning prevents biological contamination from being spun into an aerosol on the next run.

#93 · Equipment Instruments and Reagents

What is the correct action if a tube breaks inside a centrifuge during a spin?

Answer

The lid should stay closed for several minutes to allow any aerosol generated by the break to settle before opening it. Personal protective equipment should be worn while cleaning the spill, and broken glass or biohazard material must be disposed of properly. The centrifuge chamber and rotor should then be disinfected thoroughly before the next use.

#94 · Equipment Instruments and Reagents

How do you calculate the dilution factor for a simple 1 in 5 dilution?

Answer

A 1 in 5 dilution means 1 part of the original sample is mixed with enough diluent to make a total of 5 parts, so 1 part sample plus 4 parts diluent. The dilution factor here is 5. To get the true concentration of the original sample, the diluted result is multiplied back by 5.

#95 · Equipment Instruments and Reagents

If a serum sample is diluted 1 in 10 and the test reads 40 mg/dL, what is the actual concentration?

Answer

The diluted reading is multiplied by the dilution factor to find the true concentration. Here, 40 mg/dL times 10 equals 400 mg/dL. This kind of back-calculation is common whenever a sample is too concentrated to read directly on an instrument.

#96 · Equipment Instruments and Reagents

What is a standard curve, and how is it used with a spectrophotometer?

Answer

A standard curve is made by testing several samples of known concentration and plotting their absorbance readings against those known values. Once the curve is made, an unknown sample's absorbance can be compared against it to find its concentration. This method is common for older manual chemistry tests and some POCT devices.

#97 · Equipment Instruments and Reagents

Why do most lab instruments require both a low and a high level of quality control material to be run?

Answer

Running QC at two different concentration levels checks that the instrument is accurate across its full working range, not just at one point. An instrument could pass at a normal level but fail to detect a problem at an abnormal level, or the other way around. Running both levels catches problems that a single level might miss.

#98 · Equipment Instruments and Reagents

What does it mean if a quality control result falls outside the acceptable range (a QC failure)?

Answer

It means the instrument, reagent, or technique may not be producing reliable results, and patient testing should stop until the problem is found and fixed. The technologist must troubleshoot before releasing any patient results run around that time. Common causes include expired reagents, instrument malfunction, or an error in QC material preparation.

#99 · Equipment Instruments and Reagents

What is a function check, and how does it differ from full instrument calibration?

Answer

A function check is a quick daily test confirming basic parts of an instrument are working, such as a printer, timer, or temperature display. Calibration is a more detailed process that adjusts the instrument's actual measurements against known reference standards. Both are needed, but calibration happens less often than daily function checks.

#100 · Equipment Instruments and Reagents

Why should reagent bottles never be topped up by adding fresh reagent to old leftover reagent?

Answer

Mixing old and new reagent in the same bottle can mix different lot numbers together and makes it impossible to properly track expiry dates. It can also carry contamination or degraded reagent into the fresh batch. Old reagent should be used up completely or discarded before a new bottle is opened.

#101 · Equipment Instruments and Reagents

What safety equipment should be worn when preparing or handling concentrated chemical reagents?

Answer

Gloves, safety goggles or a face shield, and a lab coat should be worn to protect skin, eyes, and clothing from splashes. Work should be done in a fume hood if the reagent gives off harmful vapors. The SDS for that specific chemical should always be checked for any additional required protection.

#102 · Equipment Instruments and Reagents

What is the purpose of a fume hood, and how is it different from a biosafety cabinet?

Answer

A fume hood pulls air away from the user and vents it outside or through filtration to protect against harmful chemical vapors and fumes. A biosafety cabinet is designed to protect against biological aerosols, not chemical fumes, and often recirculates filtered air back into the room. Using the wrong one for the wrong hazard can leave the worker unprotected.

#103 · Equipment Instruments and Reagents

What does it mean for a reagent to be considered light sensitive, and how should it be stored?

Answer

A light sensitive reagent breaks down or changes chemically when exposed to light, which can make test results inaccurate. These reagents are usually stored in amber or brown bottles, or wrapped in foil, and kept away from direct light. The label will typically state to protect from light or store in the dark.

#104 · Equipment Instruments and Reagents

What routine checks should be performed on a refrigerator or freezer used to store reagents and specimens?

Answer

The temperature should be checked and logged at least once daily to confirm it stays within the required range. Door seals should be inspected for damage that could let warm air in. Any temperature excursion out of range should be documented along with the corrective action taken, such as moving reagents to a backup unit.

#105 · Equipment Instruments and Reagents

Why is it important to let a reagent or specimen reach room temperature before use if the procedure calls for it?

Answer

A cold sample or reagent can react more slowly or give inaccurate readings compared to one at the correct working temperature. Some instruments are sensitive to temperature differences between the sample and the equipment. Following the specified equilibration time in the procedure helps ensure accurate, repeatable results.

#106 · Equipment Instruments and Reagents

What is the coarse focus versus fine focus knob used for on a microscope?

Answer

The coarse focus knob moves the stage or objective a larger distance to quickly bring the specimen into general view. The fine focus knob makes small, precise adjustments to sharpen the final image once it is roughly in focus. Coarse focus is normally avoided on high power objectives because it can crash the lens into the slide.

#107 · Equipment Instruments and Reagents

What does the aperture diaphragm (iris diaphragm) on a microscope condenser control?

Answer

It controls how much light and how wide an angle of light passes up through the specimen from the condenser. Opening it too far can wash out contrast, while closing it too much can cause glare rings called diffraction artifacts. Adjusting it correctly, alongside Kohler illumination, gives the clearest and most naturally contrasted image.

#108 · Equipment Instruments and Reagents

Why must glassware and equipment used for reagent preparation be rinsed with distilled or deionized water instead of tap water?

Answer

Tap water contains minerals, chlorine, and other trace substances that can react with sensitive reagents or throw off precise chemical concentrations. Distilled or deionized water has these impurities removed, making it much more chemically neutral. Using tap water by mistake can cause inconsistent or false quality control results.

#109 · Equipment Instruments and Reagents

What should be checked on an incubator used for cultures or reagent warming before it is relied upon for testing?

Answer

Its internal temperature should be verified daily with a calibrated thermometer or built in monitor and logged to confirm it stays within the required range. The door seal should be checked to make sure it closes tightly and holds a stable temperature. Any deviation outside the acceptable range should be investigated before using the incubator for patient work.

#110 · Equipment Instruments and Reagents

Why should a technologist never rely only on an instrument's internal self-check to confirm it is working properly?

Answer

A self-check usually only confirms that internal electronics and software are functioning, not that actual test results are accurate. Real quality control material with a known expected result must still be run to verify true performance. Relying on self-checks alone can let a real testing problem go unnoticed until patient results are already affected.

#111 · Pre-Analytical Phase

What is the two-identifier rule?

Answer

Before collecting a specimen, the lab assistant must confirm at least two unique identifiers, such as full name and date of birth. This confirms the sample is linked to the right patient. Room number or bed number can never be used as an identifier because patients can move rooms.

#112 · Pre-Analytical Phase

Name three acceptable patient identifiers.

Answer

Full legal name, date of birth, and a unique health card or medical record number are all acceptable. Any two of these must match across the patient, the requisition, and the specimen label. A phone number or address is not considered a reliable identifier.

#113 · Pre-Analytical Phase

How should an inpatient be asked to state their name?

Answer

Use an open-ended question like ask the patient to state their full name and date of birth. Never ask a yes or no question such as are you John Smith, because a confused or sedated patient may just agree. Then compare the spoken answer to the requisition and wristband.

#114 · Pre-Analytical Phase

What should you do if a patient has no ID wristband?

Answer

Do not collect the specimen until a wristband is applied by nursing staff or another authorized person. Collecting without positive identification risks a misidentified sample, which can cause serious harm. Report the missing band and wait for correction before proceeding.

#115 · Pre-Analytical Phase

Why must the requisition be checked against the wristband before drawing blood?

Answer

The requisition tells you which tests are ordered and for whom, while the wristband confirms who is actually in front of you. Comparing both catches clerical mistakes, like a requisition left in the wrong room. If any detail does not match exactly, stop and resolve the discrepancy first.

#116 · Pre-Analytical Phase

What details on a requisition need to match the patient's wristband?

Answer

Full name, date of birth, and the unique identification number should all match exactly between the requisition and the wristband. Even a single letter difference in spelling needs to be investigated. This step prevents specimens from being attached to the wrong patient record.

#117 · Pre-Analytical Phase

What is a mislabeled specimen and why is it dangerous?

Answer

A mislabeled specimen is one where the label does not correctly match the patient it was drawn from. Results could be reported under the wrong patient's chart, leading to a wrong diagnosis or wrong treatment for two different people. Many labs treat this as a critical, zero tolerance error.

#118 · Pre-Analytical Phase

When must specimen labels be applied?

Answer

Labels must be applied at the bedside, in the presence of the patient, immediately after the specimen is collected. Labeling tubes ahead of time or after leaving the room increases the risk of mixing up samples between patients. This rule applies even when collecting from several patients in a row.

#119 · Pre-Analytical Phase

What minimum information must appear on a specimen label?

Answer

The label needs the patient's full name, date of birth or unique identification number, and the date and time of collection. Some facilities also require the collector's initials. This information allows the sample to be traced back to exactly one person and one collection event.

#120 · Pre-Analytical Phase

What is a wristband discrepancy?

Answer

A wristband discrepancy is any mismatch between the patient's wristband and the test requisition, such as a misspelled name or wrong date of birth. The collection should be paused until the discrepancy is resolved with nursing or registration staff. Proceeding without resolving it risks collecting from the wrong patient.

#121 · Pre-Analytical Phase

How do outpatients typically get identified before collection?

Answer

Outpatients are usually asked to state their name and date of birth, and may also show a health card or photo ID. Staff compare this spoken and printed information to the requisition. Some clinics also print a barcode wristband for outpatients to match inpatient practice.

#122 · Pre-Analytical Phase

What should happen if a patient cannot speak or is unconscious?

Answer

Identification should be confirmed using the wristband along with a second source, such as a family member, chart, or nursing staff who knows the patient. The two-identifier rule still applies even though the patient cannot answer directly. Facility policy usually outlines the exact backup steps to follow.

#123 · Pre-Analytical Phase

What is informed consent in specimen collection?

Answer

Informed consent means the patient has been told what test is being done, why it is needed, and what the procedure involves, such as a needle stick. The patient agrees to the procedure after understanding this information. Consent can be verbal for routine blood draws but should always be documented as obtained.

#124 · Pre-Analytical Phase

Does a patient have the right to refuse a blood draw?

Answer

Yes, every patient has the right to refuse any test or procedure at any time, even after the requisition was ordered. The lab assistant must respect this decision and not pressure the patient. The refusal should be reported to the ordering team or nurse and documented properly.

#125 · Pre-Analytical Phase

What should you document when a patient refuses collection?

Answer

Document the date, time, the tests that were ordered, and the reason given for refusal if the patient shared one. Notify the requesting physician or nurse so they can decide on next steps. This protects the patient's rights and creates a clear record for the care team.

#126 · Pre-Analytical Phase

Can a lab assistant collect a specimen if the patient seems hesitant but has not clearly refused?

Answer

The assistant should pause and explain the procedure again in simple terms to address the hesitation. If the patient still does not clearly agree, treat it as a refusal rather than proceeding anyway. Clear, calm communication often resolves hesitation without needing to escalate.

#127 · Pre-Analytical Phase

What is traceability in the context of specimen collection?

Answer

Traceability means every specimen can be tracked from the moment of collection through transport, processing, and testing back to one specific patient. Accurate labeling, requisition matching, and chain of custody records all support traceability. Without it, a lab cannot prove a result truly belongs to the right person.

#128 · Pre-Analytical Phase

How does a barcode system support traceability?

Answer

A barcode links the physical specimen tube to the electronic order and patient record through a unique scannable code. Scanning at each step, collection, receiving, and testing, creates a timestamped trail. This reduces manual entry errors compared to writing information by hand.

#129 · Pre-Analytical Phase

What is a chain of custody, and when is it required?

Answer

Chain of custody is a documented, signed record of everyone who handled a specimen from collection to final result, including exact times. It is required for legal or forensic samples, such as drug testing or blood alcohol tests. Any gap in the signed record can make the sample unusable as evidence.

#130 · Pre-Analytical Phase

Why should test requisition details be verified before collection, not after?

Answer

Verifying before collection lets you catch missing information, wrong tests, or wrong tube types while there is still time to fix them. Checking afterward may mean the wrong tube was already used or the wrong patient was already drawn. Prevention before the needle is inserted is always better than correcting after the fact.

#131 · Pre-Analytical Phase

What details should be verified on a test requisition?

Answer

Check the patient's identifiers, the specific tests ordered, the ordering provider, and any special instructions like fasting status. Also confirm the requisition matches the correct collection date if pre-printed labels are used. Missing or unclear information should be clarified before the patient is collected from.

#132 · Pre-Analytical Phase

What should you do if a requisition is missing the ordering physician's name?

Answer

Contact the ordering unit or use the electronic system to confirm who ordered the test before proceeding. Tests without a valid ordering provider generally cannot be billed or reported correctly. This step protects both the patient and the lab from processing an invalid order.

#133 · Pre-Analytical Phase

What does it mean to verify fasting status before collection?

Answer

Some tests, like fasting glucose or a lipid panel, require the patient to have not eaten for a set number of hours, often 8 to 12. Ask the patient directly and record the answer, since fasting status affects how results are interpreted. If the patient has eaten, the collection may need to be postponed.

#134 · Pre-Analytical Phase

What is a duplicate requisition and why does it matter?

Answer

A duplicate requisition is a second order created for the same test on the same patient at nearly the same time. Duplicates can cause unnecessary repeat draws or confusion about which sample to use. Staff should check the system to see if a valid order already exists before collecting again.

#135 · Pre-Analytical Phase

What is data privacy in relation to lab specimens and results?

Answer

Data privacy means patient information, including test results and personal identifiers, is only shared with people who are authorized to see it. This includes keeping requisitions, labels, and computer screens away from public view. Privacy protects the patient's right to confidentiality about their health information.

#136 · Pre-Analytical Phase

What law generally governs health information privacy in Canadian labs?

Answer

Provincial health information privacy legislation, along with federal privacy law, sets the rules for how personal health information is collected, used, and shared. Each province may have its own specific act, but the core principle is the same everywhere. Lab staff are trained on their facility's specific privacy policy.

#137 · Pre-Analytical Phase

How should specimen labels and requisitions be handled to protect privacy?

Answer

They should never be left visible in public areas, on countertops, or in view of other patients. Paper requisitions with patient information should be stored securely or shredded when no longer needed. Screens showing patient data should be locked or turned away from public view.

#138 · Pre-Analytical Phase

What is the correct way to dispose of paper requisitions containing patient information?

Answer

They must be placed in a designated confidential shredding bin, never in regular garbage or recycling. This prevents personal health information from being read by unauthorized people. Facilities usually have locked bins that are emptied by a secure shredding service.

#139 · Pre-Analytical Phase

Why can't old specimen labels just be thrown in the trash?

Answer

Specimen labels contain the patient's name and identifying numbers, which count as personal health information. Throwing them in regular trash risks someone finding and misusing that information. They should be disposed of through the same confidential shredding process as paper requisitions.

#140 · Pre-Analytical Phase

What should a lab assistant do if they overhear or see another patient's results by accident?

Answer

They should not repeat, discuss, or act on that information beyond what their job requires. Accidentally seeing information does not give permission to share it further. If the exposure was due to a system or process problem, it should be reported so it can be fixed.

#141 · Pre-Analytical Phase

What is a privacy breach in a lab setting?

Answer

A privacy breach happens when patient health information is seen, used, or shared by someone who is not authorized to access it. Examples include a lost requisition, an unlocked computer screen, or discussing results in a public hallway. Breaches usually must be reported right away following facility policy.

#142 · Pre-Analytical Phase

Why should a lab assistant avoid discussing patient results in hallways or elevators?

Answer

These are public areas where other patients, visitors, or staff who are not involved in the care could overhear sensitive information. Even mentioning a name with a result is a privacy violation. Patient discussions should stay in private, secure areas.

#143 · Pre-Analytical Phase

What is the purpose of verifying test request info against the electronic health record?

Answer

It confirms the printed or electronic requisition matches what the ordering provider actually entered into the system. This catches transcription errors or outdated orders that may have changed. It also helps identify if a test was already cancelled or completed.

#144 · Pre-Analytical Phase

What should you do if the test ordered on paper does not match what is in the computer system?

Answer

Stop and clarify with the ordering unit before collecting, since acting on the wrong information could mean the wrong test or no test at all. Document which source was used to resolve the conflict. Never guess which one is correct.

#145 · Pre-Analytical Phase

What is positive patient identification?

Answer

Positive patient identification is the process of confirming, without doubt, that the specimen being collected belongs to the intended patient. It relies on matching at least two identifiers using an active, not passive, method. Passive methods, like just glancing at a door sign, are not considered positive identification.

#146 · Pre-Analytical Phase

Why is asking a patient to spell their last name useful during identification?

Answer

Spelling the name out loud catches similar sounding names, such as Smith versus Smyth, that a spoken answer alone might miss. It adds an extra layer of accuracy to the verbal identification process. This is especially helpful with a long line of patients who share common first or last names.

#147 · Pre-Analytical Phase

What role does the date of birth play as an identifier?

Answer

Date of birth is a stable identifier that rarely changes and helps distinguish between two patients who might share the same name. It should always be confirmed along with the name, not used alone. A wrong or transposed date of birth is a common source of misidentification errors.

#148 · Pre-Analytical Phase

What is meant by verifying test request info for special handling instructions?

Answer

Some tests need special handling, such as being kept on ice, protected from light, or transported immediately. Checking the requisition for these notes before collection ensures the sample is handled correctly from the very first step. Missing this can cause a rejected or inaccurate result.

#149 · Pre-Analytical Phase

How should a lab assistant confirm identity for a pediatric patient?

Answer

Since a young child may not reliably answer questions, identity should be confirmed with a parent or guardian along with checking the wristband. The two-identifier rule still applies, using the wristband and guardian confirmation as the two sources. Facility policy may add extra steps for pediatric patients.

#150 · Pre-Analytical Phase

What should be done if a patient gives a nickname instead of their legal name?

Answer

Politely ask for their full legal name as it appears on their health card or identification, since records are kept under the legal name. Explain that this is needed to match their file correctly. A nickname alone should never be used to confirm identity.

#151 · Pre-Analytical Phase

Why is it important not to pre-label tubes before seeing the patient?

Answer

Pre-labeling increases the risk of putting the wrong patient's label on a tube if the order of patients changes or gets mixed up. Labels should be filled out or printed only after positive identification at the bedside. This is one of the most preventable causes of misidentification.

#152 · Pre-Analytical Phase

What is the difference between verifying a patient's identity and verifying test request information?

Answer

Verifying identity confirms who the patient actually is using identifiers like name and date of birth. Verifying test request information confirms what tests are ordered and any special requirements. Both must be done, since a correct patient with the wrong test information is still an error.

#153 · Pre-Analytical Phase

What should happen if two patients in the same room have very similar names?

Answer

Extra care should be taken to confirm the full name, date of birth, and identification number for the correct patient before collecting. Staff should never rely on bed position or a quick glance at a whiteboard. Double checking against the wristband and requisition prevents mixing up their samples.

#154 · Pre-Analytical Phase

Why is verbal confirmation alone not always enough for identification?

Answer

A patient may be confused, sedated, hard of hearing, or simply mistaken, so their spoken answer could be wrong. Verbal confirmation should always be paired with checking the wristband or another physical identifier. Relying on words alone removes an important safety check.

#155 · Pre-Analytical Phase

What is meant by data verification in the pre-analytical phase?

Answer

Data verification means checking that all the recorded information, patient identifiers, test orders, and collection details, is accurate and consistent across every document and label. This step happens before, during, and right after collection. Errors caught here prevent problems later in testing and reporting.

#156 · Pre-Analytical Phase

What should a lab assistant do if a specimen label is applied but then the patient's identity is questioned afterward?

Answer

Do not send the specimen for testing until the identity concern is fully resolved. If there is any doubt, the specimen should be discarded and recollected after identity is properly confirmed. Sending a questionable specimen risks reporting results under the wrong name.

#157 · Pre-Analytical Phase

How does confirming identity protect against wrong blood transfusions?

Answer

Blood typing and crossmatch specimens require extremely strict identification because a mismatch could cause a fatal transfusion reaction. Many facilities require a second person to independently verify identity for these samples. This extra step exists because the consequences of an error are so severe.

#158 · Pre-Analytical Phase

What is the patient's right to information before consenting to a blood draw?

Answer

Patients have the right to know what test is being performed and why, in language they can understand, before agreeing to it. This allows them to make an informed decision about whether to proceed. Skipping this explanation can undermine truly informed consent.

#159 · Pre-Analytical Phase

If a competent adult patient refuses a test ordered by their doctor, who has the final say?

Answer

The patient has the final say, since competent adults have the legal right to refuse any medical procedure. The lab assistant cannot override this decision even if a physician ordered the test. The refusal must simply be documented and communicated to the care team.

#160 · Pre-Analytical Phase

What should be checked on a requisition regarding the number and type of tubes needed?

Answer

The requisition or system should specify how many tubes and what tube types, such as a certain color top, are required for the ordered tests. Verifying this before collection avoids drawing extra unnecessary tubes or missing a required one. Using the wrong tube type can also make a specimen unsuitable for testing.

#161 · Pre-Analytical Phase

Why is it a problem to correct a specimen label by crossing out and rewriting information?

Answer

Crossed out or altered labels raise doubt about which information is actually correct and reduce confidence in the sample's identity. Most facilities require a mislabeled tube to be discarded and a new one collected and labeled properly. This keeps the chain of identification clean and trustworthy.

#162 · Pre-Analytical Phase

What is the purpose of a unique medical record number as an identifier?

Answer

A unique medical record number belongs to only one patient in the facility's system, even if names and birth dates happen to be similar between two people. It links all of a patient's records and results together reliably. Confirming this number adds an extra layer of certainty beyond name and date of birth.

#163 · Pre-Analytical Phase

What should a lab assistant do with old or unused labels left over after collection?

Answer

Any extra labels printed for a patient but not used on a specimen should be destroyed immediately, not left lying around or reused later. Leftover labels create a risk they could accidentally be applied to the wrong sample. Confidential disposal, like shredding, applies to these labels too.

#164 · Pre-Analytical Phase

How does electronic order entry help reduce identification errors compared to handwritten requisitions?

Answer

Electronic orders pull patient information directly from the registration system, reducing handwriting or transcription mistakes. Barcodes generated from electronic orders can be scanned to confirm a match instead of manually reading text. This lowers the chance of a name or number being misread.

#165 · Pre-Analytical Phase

What should a lab assistant do if a patient asks who will see their test results?

Answer

Explain that only the ordering healthcare provider and authorized care team members will have access to the results. Reassure the patient that privacy laws protect their health information from being shared elsewhere without consent. This answer supports the patient's understanding and trust in the process.

#166 · Pre-Analytical Phase

Why is proper identification described as the single most important step in specimen collection?

Answer

Every result depends on the specimen being correctly linked to the right patient, so an identification error can undo the value of the entire testing process. Even a perfectly performed collection is useless, or dangerous, if it is attached to the wrong person. This is why the two-identifier check is required every single time, with no exceptions.

#167 · Pre-Analytical Phase

What is venipuncture?

Answer

Venipuncture is the process of puncturing a vein with a needle to collect a blood sample. It is the most common way labs get blood for testing. The median cubital vein in the front of the elbow is usually the first choice because it is large and close to the skin.

#168 · Pre-Analytical Phase

Why is the median cubital vein preferred for venipuncture?

Answer

It sits close to the surface of the skin and does not move around much, which makes it easier and safer to puncture. It is also less likely to be near nerves or arteries than other veins in the arm. This lowers the chance of pain or injury during the draw.

#169 · Pre-Analytical Phase

What angle should the needle enter the skin during venipuncture?

Answer

The needle should enter the skin at an angle of about 15 to 30 degrees. A steeper angle risks going through the vein and out the other side. A flatter angle makes it harder to get into the vein at all.

#170 · Pre-Analytical Phase

How long should a tourniquet stay on the arm during a draw?

Answer

A tourniquet should not stay on for more than one minute. Leaving it on too long causes blood to pool in the area, which can change test results such as potassium or lactate levels. If more time is needed, release it and reapply after a short break.

#171 · Pre-Analytical Phase

What is the order of draw?

Answer

The order of draw is the specific sequence tubes must be filled in during a multi-tube draw. It exists to stop chemicals from one tube contaminating the next tube through the needle. Following it correctly keeps test results accurate.

#172 · Pre-Analytical Phase

List the standard order of draw from first to last tube.

Answer

The usual order is blood culture bottles first, then light blue (citrate), then serum tubes such as red or gold (SST), then green (heparin), then lavender (EDTA), and finally gray (fluoride oxalate). Each color has a different additive, and the order stops cross contamination between them.

#173 · Pre-Analytical Phase

Why do blood culture bottles go first in the order of draw?

Answer

Blood cultures need to stay as sterile as possible so bacteria from the skin do not contaminate the sample. Drawing them first avoids any additive carryover from other tubes reaching the bottle. This keeps results reliable for detecting real infections.

#174 · Pre-Analytical Phase

What additive is in a light blue top tube and what is it used for?

Answer

A light blue top tube contains sodium citrate, an anticoagulant that binds calcium to stop clotting. It is used mainly for coagulation testing such as PT and PTT. The tube must be filled completely so the blood to citrate ratio stays correct.

#175 · Pre-Analytical Phase

What additive is in a lavender top tube and what is it used for?

Answer

A lavender top tube contains EDTA, an anticoagulant that also binds calcium. It is the standard tube for a complete blood count (CBC) because it keeps the cell shapes intact. Hematology testing relies on this tube.

#176 · Pre-Analytical Phase

What additive is in a green top tube and what is it used for?

Answer

A green top tube contains heparin, an anticoagulant that works by blocking thrombin. It is commonly used for plasma chemistry tests that need faster turnaround than serum. STAT chemistry panels often use this tube.

#177 · Pre-Analytical Phase

What additive is in a gray top tube and what is it used for?

Answer

A gray top tube contains sodium fluoride and potassium oxalate. The fluoride stops glucose from breaking down in the sample, and the oxalate prevents clotting. It is the correct tube for glucose and lactate testing.

#178 · Pre-Analytical Phase

What is in a red top or gold top (SST) tube?

Answer

A plain red top tube has no additive and lets blood clot naturally to produce serum. A gold top SST tube has a clot activator plus a gel that separates serum from cells during spinning. Both are used for most routine chemistry and serology tests.

#179 · Pre-Analytical Phase

Why must anticoagulant tubes be filled to their correct volume?

Answer

Each tube is designed with a set ratio of additive to blood. If the tube is underfilled, there is too much additive compared to blood, which can change results like clotting times or cell counts. This is especially critical for the light blue citrate tube used in coagulation testing.

#180 · Pre-Analytical Phase

What does it mean to invert a tube after collection, and why do it?

Answer

Inverting means gently turning the tube upside down and back the correct number of times, not shaking it. This mixes the additive evenly through the blood. Shaking instead of inverting can damage red cells and cause hemolysis.

#181 · Pre-Analytical Phase

What is hemolysis?

Answer

Hemolysis is the breaking open of red blood cells, which releases their contents like potassium and hemoglobin into the surrounding fluid. This can falsely raise or lower certain test results. A hemolyzed sample often has to be redrawn.

#182 · Pre-Analytical Phase

Name three common causes of hemolysis during a blood draw.

Answer

Hemolysis can happen from using too small a needle for the vein, pulling the plunger back too fast on a syringe, or shaking the tube instead of gently inverting it. A tourniquet left on too long or a difficult draw with lots of probing can also cause it. Alcohol that has not fully dried on the skin before puncture is another common cause.

#183 · Pre-Analytical Phase

Why should you let alcohol dry on the skin before puncturing?

Answer

Wet alcohol entering the sample can cause hemolysis and can also sting the patient more. Waiting about 30 seconds lets the alcohol fully evaporate and finish disinfecting the site. It also helps prevent a falsely elevated result in some tests.

#184 · Pre-Analytical Phase

What is capillary collection?

Answer

Capillary collection means getting a small blood sample by puncturing the skin, usually with a lancet, instead of a vein. It draws from tiny capillaries near the skin surface rather than a full vein. It is often used for infants, small children, and point of care testing.

#185 · Pre-Analytical Phase

Where is the preferred capillary puncture site on an adult?

Answer

The palmar surface of the fingertip, usually the middle or ring finger, is the preferred site on an adult. The puncture should go across the fingerprint lines, not along them, so the drop of blood forms better. The very tip and very side of the finger should be avoided since they have less tissue.

#186 · Pre-Analytical Phase

Where is the preferred capillary puncture site on an infant, and why not the finger?

Answer

The medial or lateral plantar surface of the heel is preferred for infants. A finger stick is avoided because an infant's finger tissue is too thin and the bone is too close to the surface, risking bone infection. The heel has enough soft tissue to puncture safely.

#187 · Pre-Analytical Phase

What is the correct order of draw for capillary specimens?

Answer

For capillary collection, blood gas tubes come first, then EDTA (hematology) tubes, then other additive tubes, and serum tubes come last. This is the reverse order from venipuncture in some ways because capillary blood clots faster. Working quickly and in this order avoids clot formation ruining the sample.

#188 · Pre-Analytical Phase

Why is capillary blood collection order different from venipuncture order?

Answer

Capillary blood is exposed to air and tissue fluid right away and starts clotting much faster than blood drawn into a tube through a needle. EDTA tubes are filled early to prevent the sample from clotting before it can be collected. Serum tubes, which need clotting anyway, are filled last.

#189 · Pre-Analytical Phase

What is the first drop of blood usually discarded in a capillary puncture, and why?

Answer

The first drop is wiped away with clean gauze before collecting the sample. It may be diluted with tissue fluid or alcohol residue, which can skew results. Wiping it away helps ensure the rest of the sample is pure blood.

#190 · Pre-Analytical Phase

Why should you avoid milking or squeezing the finger too hard during capillary collection?

Answer

Excess squeezing forces tissue fluid into the blood drop, diluting the sample and causing inaccurate results. It can also cause hemolysis of the cells being squeezed out. A gentle, steady pressure is enough to keep blood flowing.

#191 · Pre-Analytical Phase

What is order of draw called when using a capillary microcollection tube system in general terms?

Answer

It follows the same underlying logic as venous order of draw, additive tubes that need mixing quickly go before tubes that clot on purpose. The Clinical and Laboratory Standards Institute (CLSI) sets this specific sequence. Following it prevents cross contamination and clotting errors in small samples.

#192 · Pre-Analytical Phase

How should a lab assistant adapt venipuncture technique for a pediatric patient?

Answer

Smaller veins and thinner skin call for a shorter, smaller gauge needle or a butterfly needle set. The child should be positioned securely, sometimes with help from a parent or a second staff member, to prevent sudden movement. Speaking calmly and explaining simply what will happen also helps reduce fear and movement.

#193 · Pre-Analytical Phase

Why is a butterfly needle often used for pediatric or difficult draws?

Answer

A butterfly needle has a short, flexible tube connecting the needle to the collection device, giving the phlebotomist more control over small or fragile veins. Its smaller needle size causes less trauma to tiny veins. The flexible tubing also allows better positioning without moving the needle in the vein.

#194 · Pre-Analytical Phase

What special care should be taken when drawing blood from an elderly patient?

Answer

Elderly patients often have thin, fragile skin and veins that roll or collapse easily, so extra care and a gentler technique are needed. Using a smaller gauge needle and less tourniquet pressure can help. Extra time may be needed to find and stabilize a suitable vein by anchoring the skin firmly.

#195 · Pre-Analytical Phase

What does it mean to anchor a vein, and why is it important in geriatric patients?

Answer

Anchoring means using your free hand to pull the skin taut below the puncture site so the vein does not move. Older patients often have veins that roll to the side, making anchoring especially important. Without anchoring, the needle can miss the vein or cause unnecessary probing.

#196 · Pre-Analytical Phase

What is a difficult draw?

Answer

A difficult draw is a venipuncture where veins are hard to find, small, rolling, fragile, or scarred, making collection harder than usual. Common causes include dehydration, obesity, chemotherapy history, or frequent past draws. Extra techniques like warming the site or trying a different arm may be needed.

#197 · Pre-Analytical Phase

How can warming the puncture site help with a difficult draw?

Answer

Applying a warm compress for a few minutes increases blood flow to the area, making veins larger and easier to see or feel. This is especially useful for patients with small or hard to find veins. It is a simple, non-invasive step to try before attempting more difficult techniques.

#198 · Pre-Analytical Phase

What should you do if you cannot find a vein after checking both arms?

Answer

You should stop and ask for help from a more experienced staff member or supervisor rather than continuing to probe the patient. Repeated failed attempts cause pain, bruising, and patient distress. Most facilities have a policy limiting the number of attempts one person can make.

#199 · Pre-Analytical Phase

What is syncope in the context of blood collection?

Answer

Syncope is fainting, a brief loss of consciousness that can happen before, during, or after a blood draw. It is often triggered by fear, pain, or the sight of blood, which lowers blood pressure and blood flow to the brain. It usually passes quickly once the patient is safely positioned.

#200 · Pre-Analytical Phase

What are early warning signs a patient may be about to faint during a draw?

Answer

Warning signs include the patient becoming pale, sweaty, or clammy, and complaining of feeling dizzy, nauseous, or lightheaded. Some patients report ringing in their ears or tunnel vision. Noticing these signs early lets staff act before the patient actually loses consciousness.

#201 · Pre-Analytical Phase

What should you do immediately if a patient shows signs of fainting during a draw?

Answer

Stop the draw right away, remove the tourniquet and needle safely, and apply pressure to the site. Lower the patient's head between their knees if seated, or lay them down flat with legs elevated if possible. Stay with the patient and call for help if needed.

#202 · Pre-Analytical Phase

Why should draws on outpatients ideally be done while the patient is seated or lying down?

Answer

Sitting or lying down reduces the distance blood has to travel back to the brain, lowering the risk of a fall if the patient faints. A standing patient who faints can be seriously injured from the fall itself. Chairs with arms also help support the patient's arm and prevent sudden movement.

#203 · Pre-Analytical Phase

What is a hematoma and how might it happen during venipuncture?

Answer

A hematoma is a bruise caused by blood leaking out of the vein into the surrounding tissue. It can happen if the needle goes through the back wall of the vein, if pressure is not applied properly after the draw, or if the patient bends their arm too soon. Applying firm pressure for at least a minute after the needle is removed helps prevent it.

#204 · Pre-Analytical Phase

How should pressure be applied to the puncture site after the draw?

Answer

Firm, steady pressure should be applied with clean gauze for at least a minute, or longer for patients on blood thinners. The arm should stay straight rather than bent, since bending can allow blood to pool under the skin. Checking the site before applying a bandage helps confirm bleeding has stopped.

#205 · Pre-Analytical Phase

What does it mean to obtain a suitable sample?

Answer

A suitable sample is one collected in the right tube, filled to the right volume, mixed properly, and free of problems like clots or hemolysis that would affect test results. It also means the sample matches the correct patient and requisition. An unsuitable sample usually must be rejected and redrawn.

#206 · Pre-Analytical Phase

Name three reasons a sample might be rejected by the lab.

Answer

Common rejection reasons include hemolysis, clots in an anticoagulant tube, insufficient volume (quantity not sufficient), and mislabeling or no label at all. Wrong tube type for the ordered test is another common cause. Any of these can force a recollection from the patient.

#207 · Pre-Analytical Phase

Why is patient identification critical before every blood draw?

Answer

Drawing from the wrong patient can lead to a serious misdiagnosis or wrong treatment based on someone else's results. Standard practice is to check two identifiers, such as full name and date of birth, against the requisition and the patient's wristband or verbal confirmation. This step must happen before any needle touches the skin.

#208 · Pre-Analytical Phase

When should tubes be labeled during a blood draw?

Answer

Tubes should be labeled at the patient's side immediately after collection, never before and never after leaving the room. Labeling in advance risks mixing up tubes between different patients. Labeling later risks forgetting or confusing which tube belongs to which person.

#209 · Pre-Analytical Phase

What information must appear on a properly labeled tube?

Answer

A properly labeled tube needs the patient's full name, date of birth or unique identification number, the date and time of collection, and the initials of the person who collected it. Facility policy may require additional details like the ordering provider. Missing any required item can cause the sample to be rejected.

#210 · Pre-Analytical Phase

What is a clotted sample and why is it a problem in an EDTA tube?

Answer

A clotted sample is one where the blood has started to solidify inside an anticoagulant tube, usually from poor or delayed mixing. In an EDTA tube used for a CBC, clots trap platelets and white cells, giving falsely low counts. The lab cannot run an accurate count on a clotted sample and must request a new one.

#211 · Pre-Analytical Phase

What does quantity not sufficient (QNS) mean?

Answer

QNS means the tube does not contain enough blood to run the requested test or to keep the correct additive ratio. It commonly happens with light blue citrate tubes that were underfilled. The lab typically rejects a QNS sample and asks for a redraw.

#212 · Pre-Analytical Phase

Why should you avoid drawing blood above an IV line in the same arm?

Answer

Fluid from the IV can mix with the blood sample and dilute it, causing inaccurate results for glucose, electrolytes, and many other tests. The preferred approach is to draw from the opposite arm whenever possible. If that arm must be used, the draw should be from a site well below the IV site, following facility protocol.

#213 · Pre-Analytical Phase

What is fist pumping during venipuncture and why is it discouraged?

Answer

Fist pumping means repeatedly clenching and unclenching the hand to make veins bulge before or during a draw. It can raise potassium levels in the blood by forcing it out of muscle cells, skewing chemistry results. Patients should just make a light fist once and hold it still.

#214 · Pre-Analytical Phase

Why should the tourniquet be released as soon as blood flow is established, or before one minute?

Answer

A tourniquet left on too long causes hemoconcentration, where fluid leaks out of the blood vessel but larger molecules and cells stay behind, concentrating the sample. This can falsely raise results for things like proteins, potassium, and cell counts. Releasing it promptly keeps the sample representative of the patient's true blood values.

#215 · Pre-Analytical Phase

What angle and direction should you enter for a fingerstick capillary puncture, relative to fingerprint ridges?

Answer

The lancet should puncture across the fingerprint ridges rather than parallel to them. This lets the blood form a round drop that is easier to collect instead of running along the ridge lines and being lost. It also produces a cleaner, faster healing wound.

#216 · Pre-Analytical Phase

What depth of puncture is generally used for a heel stick on an infant, and why does it matter?

Answer

Heel stick lancets are designed to puncture only a shallow, controlled depth, usually less than 2.4 millimeters. This is shallow enough to avoid hitting the bone (calcaneus) underneath the heel's soft tissue. Puncturing too deep risks a bone infection called osteomyelitis.

#217 · Pre-Analytical Phase

Why should you avoid areas with scarring, burns, or hematomas when choosing a puncture site?

Answer

Scar tissue and damaged skin have poor circulation, which can give an inaccurate sample or make collection harder. A hematoma from a previous draw can also cause contamination or altered results if punctured again. Choosing healthy, undamaged tissue gives the most reliable sample.

#218 · Pre-Analytical Phase

Why should a lab assistant avoid drawing from an arm on the same side as a mastectomy?

Answer

Removal of lymph nodes during mastectomy surgery can cause lymphedema and poor circulation in that arm, which can affect test accuracy and increase infection risk. It can also be more painful or difficult for the patient. The opposite arm should be used whenever available.

#219 · Pre-Analytical Phase

What should be checked on the requisition or order before beginning any blood collection?

Answer

The lab assistant should confirm the tests ordered, the correct tubes needed for those tests, any special handling like keeping a sample on ice, and the patient's identifying information. Skipping this check can lead to using the wrong tube or missing a required step. This is part of the pre-analytical phase where most lab errors happen.

#220 · Pre-Analytical Phase

Why is the pre-analytical phase considered the stage where most laboratory errors occur?

Answer

Most testing errors happen before the sample ever reaches the analyzer, during steps like patient identification, tube selection, collection technique, and handling. A mistake here, like hemolysis or mislabeling, cannot usually be fixed later in the lab. That is why careful technique during collection is so heavily emphasized.

#221 · Pre-Analytical Phase

What should you do if a patient refuses to have blood drawn?

Answer

The lab assistant should not force or pressure the patient, since consent is required for any procedure. The refusal should be reported to the nurse or supervisor and documented according to facility policy. The ordering provider may need to be informed so they can discuss it with the patient.

#222 · Pre-Analytical Phase

What general steps help prevent most collection related errors during specimen collection?

Answer

Verifying patient identity, choosing the correct tube and following order of draw, using proper technique to avoid hemolysis, and labeling at the bedside all work together to prevent errors. Staying calm and adapting technique for different patients, such as children or the elderly, also reduces failed attempts. Careful attention at each of these steps protects both the patient and the accuracy of the results.

#223 · Pre-Analytical Phase

What two pieces of information are the absolute minimum for positive patient identification at collection?

Answer

The patient must state their full legal name and date of birth out loud, not just confirm a name the collector reads to them. These are then matched against the requisition and the ID band or chart. Using only one identifier is not enough because two patients can easily share a name or a birth date.

#224 · Pre-Analytical Phase

What is a wristband or specimen label mismatch called, and why does it matter so much?

Answer

This is called a misidentification error and it is one of the most dangerous mistakes in the whole testing process. A sample labeled with the wrong name can lead to a transfusion reaction, a missed diagnosis, or treatment given to the wrong person. Most labs treat any unlabeled or mislabeled tube as an automatic reject, no exceptions.

#225 · Pre-Analytical Phase

When must a specimen tube be labeled during collection?

Answer

The tube must be labeled at the bedside or collection site, right after the draw and before leaving the patient. Labeling tubes ahead of time or after leaving the room is not allowed because it breaks the direct link between the labeled tube and the identified patient. This rule applies even if the collector is in a hurry or drawing many patients in a row.

#226 · Pre-Analytical Phase

List the core elements that must appear on a properly labeled specimen tube.

Answer

A complete label needs the patient's full name, a unique identification number, date and time of collection, and the initials of the person who collected it. Some facilities also require the collection site or a second identifier like date of birth. Missing any required element can cause the specimen to be rejected.

#227 · Pre-Analytical Phase

What does traceability mean in the pre-analytical phase?

Answer

Traceability means every specimen can be followed step by step from the patient, through collection, transport, and processing, to the final result. Each step is documented so if a problem is found later, staff can figure out exactly where and when it happened. Barcodes, timestamps, and collector initials all support this chain.

#228 · Pre-Analytical Phase

Why are barcode labels preferred over handwritten labels in most modern labs?

Answer

Barcodes reduce transcription errors because the computer reads the patient and test information directly instead of a person typing it in by hand. They also speed up accessioning since the analyzer or LIS can scan the tube and pull up orders automatically. Handwriting is still allowed as backup but is more prone to misreading.

#229 · Pre-Analytical Phase

What is chain of custody and when is it required?

Answer

Chain of custody is a strict, signed record of everyone who handled a specimen from collection to final disposal, used for legal or forensic samples like drug testing or blood alcohol cases. Every transfer of the specimen must be documented with a signature, date, and time, with no gaps allowed. If the chain is broken, the sample may not be accepted as evidence in court.

#230 · Pre-Analytical Phase

What temperature range counts as room temperature storage for most specimens?

Answer

Room temperature is generally considered about 18 to 25 degrees Celsius. Many chemistry and hematology specimens are stable at this range for a limited number of hours after collection. Specific stability times still depend on the test, so the lab's procedure manual should always be checked.

#231 · Pre-Analytical Phase

What temperature range is used for refrigerated specimen storage?

Answer

Refrigerated storage is usually 2 to 8 degrees Celsius. Many specimens that are not tested right away, such as certain chemistry or microbiology samples, are kept at this temperature to slow down chemical breakdown and bacterial growth. Refrigeration is gentler than freezing and preserves most analytes without damaging cells as much.

#232 · Pre-Analytical Phase

Why do some tests, like ammonia and lactic acid, require the specimen to be transported on ice?

Answer

Ammonia and lactic acid levels rise quickly at room temperature because ongoing cell metabolism keeps producing them after the blood leaves the body. Chilling the specimen on wet ice slows this metabolic activity down so the result reflects the patient's true level. If these specimens sit out too long, the result will be falsely elevated.

#233 · Pre-Analytical Phase

What is the difference between wet ice and dry ice for specimen transport?

Answer

Wet ice is frozen water and keeps a specimen cold, around 0 to 4 degrees Celsius, without freezing it solid. Dry ice is frozen carbon dioxide and is much colder, around minus 78 degrees Celsius, so it is used to keep already-frozen specimens frozen during shipping. Using dry ice on a specimen that should stay only chilled, not frozen, can ruin the sample.

#234 · Pre-Analytical Phase

Why must dry ice shipments never be sealed in a completely airtight container?

Answer

Dry ice slowly turns from solid straight into carbon dioxide gas, a process called sublimation. If the gas has no way to escape, pressure builds up inside the container and it can burst or explode. Shipping containers are designed with venting to let this gas release safely.

#235 · Pre-Analytical Phase

What is liquid nitrogen used for in specimen handling, and what safety concern comes with it?

Answer

Liquid nitrogen is used to flash freeze or store specimens at extremely low temperatures, often below minus 150 degrees Celsius, for things like certain molecular or research samples. It can cause severe frostbite on contact with skin within seconds. Staff handling it need insulated gloves, a face shield, and good ventilation since it can also displace oxygen in a closed room.

#236 · Pre-Analytical Phase

What does TDG stand for and why does it matter to a lab assistant shipping specimens?

Answer

TDG stands for Transportation of Dangerous Goods, the Canadian federal regulation covering how hazardous materials, including biological and diagnostic specimens, must be packaged and labeled for shipment. Lab assistants who prepare specimens for courier or postal shipping need basic TDG awareness training. This ensures specimens are triple packaged, labeled correctly, and handled safely if there is a leak or accident in transit.

#237 · Pre-Analytical Phase

Describe the triple packaging system required for shipping most diagnostic specimens.

Answer

The system has three layers, a primary leakproof container holding the actual specimen, a secondary leakproof container with absorbent material around the primary container, and a rigid outer shipping box. This way, if the inner container leaks, the absorbent material and the secondary container catch it before anything reaches the outside. This protects couriers, postal workers, and anyone else who touches the package.

#238 · Pre-Analytical Phase

What is a Category B biological substance, and how is it labeled for shipping?

Answer

Category B covers infectious substances that could cause disease in humans or animals but are not likely to cause permanent disability or death, which describes most routine diagnostic specimens. These packages are marked with the UN3373 diamond-shaped label. It signals to handlers that the contents need care but are lower risk than Category A infectious substances.

#239 · Pre-Analytical Phase

Why should specimen tubes always be transported upright rather than lying flat, when possible?

Answer

Keeping tubes upright helps clot formation in serum tubes settle properly and reduces the chance of hemolysis from the specimen sloshing around. It also lowers the risk of leakage around the stopper. Most transport racks and bags are designed specifically to hold tubes vertically.

#240 · Pre-Analytical Phase

What is a pneumatic tube system, and what specimen quality risk does it carry?

Answer

A pneumatic tube system uses air pressure to shoot carriers containing specimens quickly through tubes between a nursing unit and the lab. The sudden acceleration, deceleration, and jostling can cause hemolysis or platelet clumping in fragile specimens. Some tests, like certain coagulation studies, may be restricted from pneumatic transport and require hand delivery instead.

#241 · Pre-Analytical Phase

What is hemolysis and how can rough handling during transport cause it?

Answer

Hemolysis is the rupture of red blood cells, which releases hemoglobin and other cell contents into the surrounding plasma or serum. Shaking a tube vigorously, dropping it, or sending it through a rough pneumatic tube system can physically break the cells open. A hemolyzed specimen can falsely raise potassium, LDH, and other results, often forcing a recollection.

#242 · Pre-Analytical Phase

What is the general rule for how long a whole blood specimen can sit before the serum or plasma is separated from the cells?

Answer

Most labs require separation within one to two hours of collection, though the exact time depends on the test and facility protocol. Leaving cells in contact with serum or plasma too long allows ongoing metabolism and leakage that changes analyte levels, such as falsely lowering glucose or raising potassium. Timely centrifugation and separation protect specimen integrity until testing can occur.

#243 · Pre-Analytical Phase

A tube arrives at the lab with no time of collection on the label. What should the lab assistant do?

Answer

The specimen should be rejected or held pending clarification rather than guessed at, since missing collection time can affect analyte stability and turnaround decisions. The assistant should contact the collecting unit to find out if the missing information can be verified and documented before proceeding. Never estimate the time or make one up to fill the gap.

#244 · Pre-Analytical Phase

What is accessioning in the laboratory information system, and when does it happen?

Answer

Accessioning is the process of logging a specimen into the LIS, assigning it a unique accession number, and linking it to the correct patient and ordered tests. This happens as soon as the specimen and requisition arrive at the lab, before any testing begins. The accession number becomes the specimen's tracking identity throughout the rest of its journey.

#245 · Pre-Analytical Phase

Why is a unique accession number assigned to every specimen instead of just using the patient's name?

Answer

Names can be duplicated between patients and are not a reliable way to track a single tube through many processing steps. A unique accession number lets the LIS, the analyzer, and every staff member reference one specific specimen without confusion. It also becomes the link between the physical tube and the electronic record for results.

#246 · Pre-Analytical Phase

What should happen if a requisition and specimen label do not match during accessioning?

Answer

The specimen should not be accepted for testing until the discrepancy is resolved, because processing it could attach a result to the wrong patient. The lab assistant should hold the specimen, document what does not match, and contact the ordering unit or collector for clarification. This is treated as a critical error, not a minor clerical issue.

#247 · Pre-Analytical Phase

Give three examples of common pre-analytical errors besides mislabeling.

Answer

Common examples include using the wrong tube type or additive for the ordered test, underfilling a tube so the blood to additive ratio is off, and prolonged tourniquet application causing hemoconcentration. Collecting from an arm with an active IV line is another frequent error that can dilute or contaminate results. Each of these can distort results even though the patient and label are perfectly correct.

#248 · Pre-Analytical Phase

What is a short draw, and why is it a rejection criterion for coagulation testing?

Answer

A short draw happens when a tube, especially a citrate tube for coagulation tests, is not filled to its designated volume. Citrate tubes need a precise blood to anticontaminant ratio because the citrate binds calcium to prevent clotting, and too little blood means too much citrate relative to the sample. This throws off PT and PTT results, so most labs automatically reject citrate tubes filled below about 90 percent.

#249 · Pre-Analytical Phase

What course of action should a lab assistant take when a specimen is found to be grossly hemolyzed on arrival?

Answer

The assistant should check whether the ordered test is affected by hemolysis and reject the specimen if it is, following the lab's rejection criteria. The rejection and reason should be documented, and the collecting unit notified so a new specimen can be drawn. Results should never be released with a note hoping the clinician will interpret it correctly if the test truly cannot be run on a hemolyzed sample.

#250 · Pre-Analytical Phase

Why is prolonged tourniquet application, over about one minute, considered a pre-analytical error?

Answer

Leaving a tourniquet on too long causes hemoconcentration, where fluid shifts out of the blood vessel and the remaining blood becomes more concentrated with cells and large molecules. This can falsely elevate results like potassium, protein, and cell counts. Tourniquets should be released within about one minute of application, ideally as soon as blood flow starts.

#251 · Pre-Analytical Phase

How can contamination from alcohol swabs affect a blood alcohol specimen, and how is it prevented?

Answer

Alcohol left on the skin from a swab can seep into the puncture site and falsely raise a blood alcohol result. For blood alcohol collections, the site is cleaned with a non-alcohol antiseptic such as benzalkonium chloride or povidone-iodine instead. This avoids any chance the specimen itself picks up outside alcohol contamination.

#252 · Pre-Analytical Phase

What does order of draw refer to, and why does it matter for preventing contamination?

Answer

Order of draw is the required sequence for filling multiple tubes during one venipuncture, generally blood cultures first, then plain or clot tubes, then citrate, then heparin, then EDTA, then fluoride. Following this sequence prevents additive carryover between tubes, where residue from one tube's stopper transfers into the next tube's contents. Getting the order wrong can contaminate a tube with the wrong additive and distort results.

#253 · Pre-Analytical Phase

Why must a lab assistant never draw a blood culture through an existing IV line?

Answer

IV lines and the skin around them are more likely to harbor bacteria from the catheter or infusion site, which can contaminate the culture and produce a false positive result. A false positive can lead to unnecessary antibiotics and a longer hospital stay for the patient. Blood cultures should be collected by fresh venipuncture using strict aseptic technique instead.

#254 · Pre-Analytical Phase

What steps reduce skin contamination risk when collecting a blood culture?

Answer

The site is cleaned with a proper antiseptic, such as chlorhexidine, using a scrubbing motion and allowed to air dry completely before the needle goes in. The site should not be touched or repalpated after cleaning unless the collector is wearing sterile gloves. These steps reduce the chance that normal skin bacteria end up in the culture bottle and are mistaken for a true infection.

#255 · Pre-Analytical Phase

What is specimen stability, and why does it matter for how quickly a sample must reach the lab?

Answer

Specimen stability is the length of time an analyte stays accurate and unchanged in a specimen after collection, under specified storage conditions. Some analytes, like glucose or blood gases, change quickly and need almost immediate testing or special handling, while others like a basic chemistry panel are stable for hours. Knowing stability limits tells staff how urgently a specimen needs to move from collection to processing.

#256 · Pre-Analytical Phase

Why are blood gas specimens handled differently in terms of speed and temperature compared to routine chemistry?

Answer

Blood gas results, like oxygen and carbon dioxide levels, change within minutes because red and white cells keep consuming oxygen and producing carbon dioxide after the draw. These specimens are placed on ice immediately and rushed to the lab, often run within 15 to 30 minutes of collection. Delayed or unchilled blood gas specimens will not reflect the patient's true respiratory or metabolic status.

#257 · Pre-Analytical Phase

What does STAT mean in the context of specimen priority, and how should a lab assistant treat a STAT order differently?

Answer

STAT means the test is urgently needed, often because the patient's condition is critical or changing quickly. A STAT specimen should be collected, transported, accessioned, and processed ahead of routine specimens, skipping the normal queue wherever safely possible. Delaying a STAT specimen can directly affect urgent patient care decisions.

#258 · Pre-Analytical Phase

What should a lab assistant check first when a batch of specimens arrives from a long-distance courier run?

Answer

The assistant should check that the transport temperature was maintained, usually by looking at an included temperature log or cold pack condition, and confirm all tubes are intact with no leaks or broken caps. Any specimen that traveled outside its required temperature range or arrived compromised should be flagged for possible rejection. This initial check happens before specimens are logged in as acceptable for testing.

#259 · Pre-Analytical Phase

Why do 24 hour urine collections need to be refrigerated or kept on ice throughout the collection period?

Answer

Urine left at room temperature for a full day allows bacterial growth and chemical breakdown that can change analyte levels like glucose or certain hormones. Refrigeration slows this bacterial activity and helps preserve the sample's original composition. Patients are usually given a container to keep in the refrigerator for the entire collection period.

#260 · Pre-Analytical Phase

What is the purpose of a specimen collection manual, and who is expected to follow it?

Answer

A specimen collection manual lists exact requirements for every test the lab offers, including tube type, volume, storage temperature, and stability window. Every person who collects or handles specimens, including nurses, phlebotomists, and lab assistants, is expected to follow it consistently. Following the manual keeps results reliable no matter who collected the specimen or where.

#261 · Pre-Analytical Phase

Why should a lab assistant never re-cap a used needle before disposal, and how does this connect to specimen safety?

Answer

Re-capping a used needle by hand greatly increases the risk of an accidental needlestick injury to the person handling it. Standard practice is to drop the needle directly into a sharps container immediately after use, uncapped. This protects the collector's safety while the specimen itself is being properly labeled and transported.

#262 · Pre-Analytical Phase

What information should a chain of custody form capture at every transfer point?

Answer

Each transfer needs the date and time, the printed name and signature of the person releasing the specimen, and the printed name and signature of the person receiving it. Any gap or missing signature breaks the documented chain and can make the specimen unusable as legal evidence. This is stricter than the documentation required for a routine clinical specimen.

#263 · Pre-Analytical Phase

How should specimens for infectious disease testing, like suspected tuberculosis, be labeled and handled differently during transport?

Answer

These specimens are usually marked with a biohazard label so everyone who handles them knows to use extra caution and appropriate personal protective equipment. They may also require specific containment, like being sealed in a separate bag from other specimens. This labeling protects lab staff and couriers from unnecessary exposure risk.

#264 · Pre-Analytical Phase

What is the difference between a specimen rejection and a specimen recollection request?

Answer

Rejection is the lab's decision not to test a specimen because it fails to meet quality criteria, such as being hemolyzed, clotted, or mislabeled. Recollection is the follow up action, asking the collecting unit to draw a brand new specimen from the patient to replace the rejected one. Both steps are documented so there is a clear record of why the original result was delayed.

#265 · Pre-Analytical Phase

Why can a clotted specimen in an EDTA tube not be used for a complete blood count?

Answer

EDTA prevents clotting by binding calcium, but if the tube is under filled, mixed poorly, or the additive fails, small clots can form in the sample. A clot means red cells, white cells, and platelets are trapped unevenly, so an automated cell counter would give inaccurate counts. Any visible clot in an EDTA tube means the specimen must be rejected and redrawn.

#266 · Pre-Analytical Phase

What does it mean to invert a tube after collection, and why is gentle inversion important rather than shaking?

Answer

Inverting means turning the tube upside down and back a set number of times, usually 5 to 10, to mix the blood with the additive inside. This must be done gently because vigorous shaking can cause hemolysis or foam formation. Proper inversion ensures the additive works evenly throughout the sample without damaging the cells.

#267 · Pre-Analytical Phase

Why do some analytes, like potassium, need special attention regarding hemolysis specifically?

Answer

Potassium is present in much higher concentration inside red blood cells than in the surrounding plasma. When cells rupture during hemolysis, that intracellular potassium spills out into the plasma and falsely raises the measured level. Even mild hemolysis can be enough to push a potassium result outside its true range.

#268 · Pre-Analytical Phase

What is the recommended action if a specimen must be transported over a long distance and will take several hours to reach the lab?

Answer

The specimen should be packed according to its specific stability requirements, which might mean adding cold packs, using an insulated container, or separating serum from cells before shipping if that is standard practice. The goal is to keep the analyte as close to its collection-time value as possible for the whole trip. Facilities usually have a written protocol for exactly how to prepare specimens for extended transport.

#269 · Pre-Analytical Phase

Why is light exposure a concern for certain specimens, such as bilirubin or vitamin levels?

Answer

Some analytes, including bilirubin and certain vitamins, break down when exposed to light, which can falsely lower the measured result. These specimens are usually protected by wrapping the tube in foil or using an amber colored collection tube. Leaving a light-sensitive specimen sitting in a bright room before testing can give a misleadingly low result.

#270 · Pre-Analytical Phase

What should a lab assistant do if a specimen arrives correctly labeled but the requisition is missing entirely?

Answer

The specimen should not be processed until a valid requisition or verified electronic order is obtained, since testing without an order means there is no record of what was actually requested or why. The assistant should contact the ordering department to get the missing paperwork or confirm the order electronically. This protects against testing errors and billing problems down the line.

#271 · Pre-Analytical Phase

Why are some coagulation specimens restricted from pneumatic tube transport even when other specimens are allowed?

Answer

Coagulation testing, like PT and PTT, is sensitive to any mechanical disruption of platelets and clotting factors, which the sudden forces in a pneumatic tube system can trigger prematurely. This can cause falsely abnormal coagulation results that do not reflect the patient's real clotting status. Many facilities require these specimens to be hand carried instead.

#272 · Pre-Analytical Phase

What is the purpose of a specimen tracking log separate from the LIS, and when might one still be used?

Answer

A manual or backup tracking log records specimen movement when the electronic system is down or as an extra layer of accountability for high risk specimens like legal or blood bank samples. It typically lists the accession number, collection time, and who handled the specimen at each step. This backup ensures traceability is never fully lost even during a computer outage.

#273 · Pre-Analytical Phase

What is the general rule about the maximum time a whole blood specimen destined for a blood bank crossmatch can be stored before it is considered too old to use?

Answer

Many facilities require blood bank samples to be no more than 3 days old at the time of transfusion if the patient has had a transfusion or been pregnant in the last 3 months, because new antibodies could have formed. This rule protects against a crossmatch being done on outdated antibody information. Blood bank specimens are handled with extra strict labeling and expiration rules compared to routine chemistry samples.

#274 · Pre-Analytical Phase

Why should specimens never be left at room temperature on a counter for an extended period while waiting to be picked up?

Answer

Leaving specimens out ignores whatever stability window applies to the ordered tests, and cells continue to metabolize glucose, release potassium, and break down other analytes the longer they sit. This can produce results that no longer reflect the patient's actual condition. Specimens should be stored per protocol, whether that is room temperature, refrigerated, or on ice, immediately after collection while awaiting transport.

#275 · Pre-Analytical Phase

What is the difference between accessioning and specimen collection in terms of who typically performs each step?

Answer

Specimen collection is usually done by a phlebotomist, nurse, or lab assistant directly with the patient, drawing and labeling the tube at the bedside. Accessioning happens afterward at the laboratory, usually by a lab assistant or technologist, entering the specimen into the LIS and assigning it an accession number. Both steps are essential parts of the same traceability chain but happen in different locations and by potentially different staff.

#276 · Pre-Analytical Phase

Why is it important to minimize the time a tourniquet is applied and to release it before removing the needle?

Answer

Releasing the tourniquet before withdrawing the needle reduces bruising and hematoma formation at the puncture site. It also prevents ongoing hemoconcentration from continuing right up until the last drop of blood is drawn. Best practice is to release the tourniquet as soon as adequate blood flow is confirmed, generally within about one minute of application.

#277 · Pre-Analytical Phase

What action should be taken if a specimen container is found to be leaking when a courier delivers a batch to the lab?

Answer

The leaking container should be handled with gloves and set aside following the lab's spill and biohazard procedure, and the specimen inside is generally rejected since contamination and volume loss cannot be ruled out. The event should be documented and reported, including notifying the sending unit that a new specimen is needed. The rest of the batch should be checked to make sure no cross contamination occurred from the leak.

#278 · Pre-Analytical Phase

Why do frozen specimens need to be transported on dry ice rather than simply placed in a regular cooler with ice packs?

Answer

Regular ice packs only keep a specimen cold, typically a few degrees above freezing, but cannot keep an already-frozen specimen solidly frozen during a long trip. Dry ice maintains temperatures far below freezing, so a frozen specimen stays frozen the entire time instead of partially thawing and refreezing, which can damage the sample. Partial thawing can degrade sensitive analytes like certain hormones or molecular targets.

#279 · Pre-Analytical Phase

Summarize the main goal that ties together labeling, storage temperature, transport method, and chain of custody rules in the pre-analytical phase.

Answer

All of these rules exist to make sure the specimen that reaches the analyzer is still an accurate, untampered representation of what came out of the patient. Every step, from correct labeling to proper temperature control to documented handoffs, protects that link between patient and result. If any one of these steps fails, the final test result cannot be trusted, no matter how well the actual testing is performed.

#280 · Pre-Analytical Phase

What does it mean to assess specimen suitability before testing?

Answer

It means checking that the sample was collected, labeled, and stored correctly before any testing begins. The lab assistant looks for the right tube type, enough volume, correct clotting or anticoagulation, and no visible damage. A sample that fails these checks should be rejected rather than tested.

#281 · Pre-Analytical Phase

What is hemolysis and why does it matter?

Answer

Hemolysis is the breaking open of red blood cells, which releases their contents into the plasma or serum. It turns the sample pink or red and can falsely raise results like potassium, LDH, and AST. Many hemolyzed samples must be rejected and redrawn.

#282 · Pre-Analytical Phase

What is lipemia in a blood sample?

Answer

Lipemia is a milky or cloudy appearance in serum or plasma caused by high levels of fat particles, usually from a patient who recently ate a fatty meal. It can interfere with light-based testing methods and skew results. Some labs need the patient to fast before redrawing.

#283 · Pre-Analytical Phase

What is icterus in a specimen?

Answer

Icterus is a yellow or amber discoloration of serum or plasma caused by high bilirubin levels. It can happen with liver disease or bile duct blockage. Like hemolysis and lipemia, it can interfere with certain chemistry tests and should be noted.

#284 · Pre-Analytical Phase

Why must a clotted sample be rejected when anticoagulated tubes are required?

Answer

Anticoagulant tubes like EDTA or citrate need liquid whole blood to give accurate cell counts or coagulation results. A clot means the anticoagulant did not mix properly or failed, so cells are trapped and counts will be wrong. The safest action is to reject the sample and request a new draw.

#285 · Pre-Analytical Phase

What does QNS mean on a rejected specimen?

Answer

QNS stands for quantity not sufficient. It means the tube does not contain enough blood or fluid to run the requested test accurately. This is common with underfilled tubes, which can also throw off the additive-to-blood ratio.

#286 · Pre-Analytical Phase

Why is the additive-to-blood ratio important in tubes like citrate for coagulation testing?

Answer

Coagulation tubes need a precise ratio of blood to anticoagulant, usually nine parts blood to one part citrate. An underfilled tube has too much anticoagulant relative to blood, which falsely prolongs clotting times like PT and PTT. This is why partially filled coag tubes are rejected.

#287 · Pre-Analytical Phase

What should a lab assistant check on the specimen label before processing?

Answer

The label should match the requisition exactly, including patient name, date of birth, and a unique identifier like a medical record number. Time and date of collection and collector initials should also be present. Mismatched or missing labels are a major cause of specimen rejection.

#288 · Pre-Analytical Phase

What is aliquoting?

Answer

Aliquoting means dividing a larger specimen into smaller labeled portions so multiple tests or departments can use the same original sample. Each aliquot must be labeled with the same patient information as the original tube. This protects the primary sample from repeated freeze-thaw or contamination.

#289 · Pre-Analytical Phase

Why do labs aliquot serum or plasma into secondary tubes instead of sending the original tube everywhere?

Answer

Sending the original tube to multiple stations risks mislabeling, spills, and running low on volume. Aliquoting keeps the primary tube safe as a backup for repeat testing. It also lets different instruments or departments work on the sample at the same time.

#290 · Pre-Analytical Phase

What is a dilution in laboratory terms?

Answer

A dilution is mixing a sample with a diluent, such as saline, to reduce its concentration before testing. This is often done when a result would otherwise be too high for the instrument to measure accurately. The final result must then be multiplied by the dilution factor.

#291 · Pre-Analytical Phase

How do you calculate a simple dilution factor, like a 1:5 dilution?

Answer

A 1:5 dilution means one part sample plus four parts diluent, for a total of five parts. The dilution factor is the total parts divided by the sample parts, so here it is 5. Any result from that diluted sample must be multiplied by 5 to get the true value.

#292 · Pre-Analytical Phase

What is a serial dilution?

Answer

A serial dilution is a series of step by step dilutions, where each tube is diluted from the tube before it rather than from the original sample. This is useful for finding antibody titers or bacterial counts across a wide range. Each step usually uses the same dilution ratio, like 1:2 each time.

#293 · Pre-Analytical Phase

Why must diluent and sample be mixed thoroughly before testing a diluted specimen?

Answer

If the diluent and sample are not mixed well, the sample will not be evenly distributed throughout the tube. This causes inaccurate and inconsistent results depending on where the aliquot is taken from. Gentle inversion or vortexing ensures a uniform mixture.

#294 · Pre-Analytical Phase

What is the basic purpose of culturing a specimen in microbiology?

Answer

Culturing means placing a specimen onto or into growth media so that any bacteria or other organisms present can multiply into visible colonies. This lets the lab identify what organism is causing an infection. Without growth, many organisms cannot be identified or tested for antibiotic sensitivity.

#295 · Pre-Analytical Phase

What is inoculation in microbiology?

Answer

Inoculation is the process of transferring a small amount of specimen onto a culture plate or into broth using a sterile loop or swab. It introduces the organism to an environment where it can grow. Careful technique prevents contamination from outside organisms.

#296 · Pre-Analytical Phase

What is the streak plate technique used for?

Answer

The streak plate technique spreads a specimen thinly across an agar plate in a pattern of decreasing density, usually in three or four sections. This separates individual organisms so they grow into distinct, countable colonies. It is the standard method for isolating pure cultures from a mixed specimen.

#297 · Pre-Analytical Phase

What is the difference between selective and differential culture media?

Answer

Selective media contains substances that allow only certain organisms to grow while inhibiting others. Differential media contains indicators that make different organisms look different, such as changing color, even if many types can grow. Some media, like MacConkey agar, do both jobs at once.

#298 · Pre-Analytical Phase

Why must culture plates be incubated at a specific temperature and atmosphere?

Answer

Most human pathogens grow best near body temperature, around 35 to 37 degrees Celsius. Some organisms also need specific gas conditions, such as increased carbon dioxide or no oxygen at all, to grow properly. Using the wrong temperature or atmosphere can cause false negative results.

#299 · Pre-Analytical Phase

What is the basic goal of DNA extraction from a specimen?

Answer

DNA extraction pulls genetic material out of cells and separates it from proteins, lipids, and other cell debris. This usually involves breaking open the cell, removing proteins, and then purifying and concentrating the DNA. Clean DNA is needed for accurate molecular testing like PCR.

#300 · Pre-Analytical Phase

What is the basic goal of RNA extraction, and why is it more delicate than DNA extraction?

Answer

RNA extraction isolates ribonucleic acid from a sample so it can be used in molecular tests, similarly to DNA extraction. RNA is more fragile because it is single stranded and enzymes called RNases, found almost everywhere including on skin, break it down quickly. Specimens for RNA testing need to be handled carefully and kept cold to avoid degradation.

#301 · Pre-Analytical Phase

Why is cell lysis the first major step in nucleic acid extraction?

Answer

Lysis breaks open the cell membrane and, if present, the cell wall, so that the DNA or RNA inside can be released. This can be done using detergents, enzymes, or mechanical methods depending on the sample type. Without lysis, the genetic material stays trapped inside the cell and cannot be purified.

#302 · Pre-Analytical Phase

What does it mean to make a blood smear using the wedge method?

Answer

The wedge method uses a spreader slide held at an angle to pull a drop of blood across a second slide, creating a thin film with a feathered edge. A good smear has a smooth gradient from thick to thin with no streaks or holes. This feathered edge is where cells are best spread out for accurate identification.

#303 · Pre-Analytical Phase

What angle is typically used for the spreader slide when making a manual blood smear?

Answer

The spreader slide is usually held at a 30 to 45 degree angle to the slide holding the blood drop. A steeper angle makes a shorter, thicker smear, while a lower angle makes a longer, thinner smear. The angle should be adjusted based on how thick or watery the blood sample is.

#304 · Pre-Analytical Phase

What are common defects seen on a poorly made blood smear?

Answer

Common defects include holes or vacuoles from grease or dust on the slide, streaks from a chipped spreader slide edge, and a smear that ends too abruptly or too thin. Uneven pressure or hesitation while spreading also causes ridges. Any of these defects can distort cell shapes and make evaluation inaccurate.

#305 · Pre-Analytical Phase

Why must a blood smear be air dried quickly after it is made?

Answer

If a smear dries too slowly, red blood cells can shrink, crenate, or change shape before they are fixed. Fanning the slide or using a fast fan helps it dry within a few seconds. A properly and quickly dried smear preserves normal cell morphology for staining.

#306 · Pre-Analytical Phase

What does fixing a smear mean, and why is methanol commonly used?

Answer

Fixing means treating the dried smear with a chemical, usually methanol, that denatures and preserves the proteins and cell structures so cells do not wash away or distort during staining. Methanol also kills any remaining organisms, making the slide safer to handle. Fixation must happen before most staining procedures begin.

#307 · Pre-Analytical Phase

What is the general principle of the Gram stain?

Answer

The Gram stain uses crystal violet, iodine, a decolorizer like alcohol, and a safranin counterstain to separate bacteria into two groups based on their cell wall structure. Gram positive bacteria have a thick peptidoglycan wall that traps the crystal violet, staining them purple. Gram negative bacteria have a thinner wall that loses the purple stain and picks up the pink safranin counterstain instead.

#308 · Pre-Analytical Phase

In the Gram stain procedure, what is the role of the iodine step?

Answer

Iodine acts as a mordant, meaning it binds to the crystal violet already inside the cell and forms a larger complex. This complex is too big to easily wash out of gram positive cell walls during decolorization. Without iodine, both types of bacteria would lose the purple stain.

#309 · Pre-Analytical Phase

What happens if the decolorization step in a Gram stain is left on too long?

Answer

Over decolorizing can strip the crystal violet even out of gram positive organisms, making them falsely appear gram negative. This is one of the most common technical errors in Gram staining. Correct timing and technique are critical for an accurate result.

#310 · Pre-Analytical Phase

What is the Wright stain used for?

Answer

The Wright stain is a Romanowsky type stain used to color blood smears so that red cells, white cells, and platelets can be told apart under a microscope. It contains a mix of acidic eosin and basic methylene blue dyes. Different cell structures pick up different amounts of each dye, creating a range of colors that help identify cell types.

#311 · Pre-Analytical Phase

What is the difference between the Wright stain and the Jenner-Giemsa stain?

Answer

Both are Romanowsky type stains used on blood and bone marrow smears, and both use combinations of eosin and methylene blue derivatives. Jenner-Giemsa often gives more detailed staining of nuclear chromatin patterns and is favored in some labs for reviewing subtle nuclear detail. In practice, they are used for the same general purpose but may differ slightly in exact dye ratios and staining time.

#312 · Pre-Analytical Phase

What is the Leishman stain used for?

Answer

The Leishman stain is another Romanowsky type stain used mainly to examine blood films for blood cell morphology and to help identify parasites like malaria. It is prepared in methanol, which fixes the smear at the same time it stains it. It is common in some countries as an alternative to Wright stain.

#313 · Pre-Analytical Phase

What is H&E; stain and where is it used?

Answer

H&E; stands for hematoxylin and eosin, and it is the standard stain used in histology to look at tissue sections rather than blood smears. Hematoxylin stains the nucleus of cells a dark blue or purple, while eosin stains the cytoplasm and other structures pink. This combination lets pathologists clearly see tissue architecture and cell detail.

#314 · Pre-Analytical Phase

What is the Papanicolaou, or Pap, stain used for?

Answer

The Papanicolaou stain is used mainly on cervical and other cytology specimens to detect abnormal or cancerous cells. It uses multiple dyes to give a range of colors that highlight differences between the nucleus and cytoplasm. This detail helps cytotechnologists spot changes in cell shape and staining pattern that suggest disease.

#315 · Pre-Analytical Phase

Why must Pap smear slides usually be fixed immediately with a wet fixative rather than air dried?

Answer

Cervical cells air dry very quickly and can shrink or distort within seconds, which destroys the fine detail needed to detect abnormal cells. A wet fixative, often an alcohol spray or bath, is applied immediately after the smear is made to preserve cell structure. Air drying artifact is one of the most common reasons a Pap smear is rejected as unsatisfactory.

#316 · Pre-Analytical Phase

What do all Romanowsky stains, like Wright, Giemsa, and Leishman, have in common?

Answer

All Romanowsky stains combine an acidic dye, eosin, with a basic dye related to methylene blue. Acidic cell structures attract the basic dye and appear blue or purple, while basic structures attract the acidic dye and appear pink or red. This dual staining is what allows different white blood cell types to be told apart.

#317 · Pre-Analytical Phase

What does 'loading equipment' generally involve for a lab assistant working with automated analyzers?

Answer

Loading equipment means placing labeled specimen tubes into the correct racks or carousels in the order and orientation the analyzer requires. It also includes checking that barcodes face the reader and that tubes are properly capped or uncapped as needed. Incorrect loading can cause the instrument to misread or skip a sample.

#318 · Pre-Analytical Phase

Why is proper centrifuge loading important before analyzers can use a specimen?

Answer

Tubes must be balanced across from each other in the centrifuge with tubes of similar weight, or the machine can vibrate, damage the rotor, or fail to separate the sample properly. Centrifuging separates serum or plasma from cells so the liquid portion can be tested. An unbalanced or improperly capped tube can also leak or hemolyze during the spin.

#319 · Pre-Analytical Phase

Why do many analyzers require a minimum dead volume in a tube beyond what is needed for the test?

Answer

Dead volume is the extra sample left in the tube that the instrument's probe cannot reach or draw from without risking a clog or air bubble. If the tube is filled with barely enough for the test, the analyzer may fail to aspirate properly. Lab assistants should check the specific dead volume requirement before submitting a low volume sample.

#320 · Pre-Analytical Phase

What is a biologic variable in the context of pre-analytical testing?

Answer

A biologic variable is a normal, natural factor about the patient or their body that can change test results even when technique and equipment are perfect. Examples include age, sex, diet, exercise, and time of day. These are different from lab errors because they reflect true biological differences rather than mistakes.

#321 · Pre-Analytical Phase

How can diurnal variation affect certain lab results?

Answer

Diurnal variation is the natural rise and fall of certain substances at different times of day, such as cortisol being highest in the morning and lowest at night. If a test is meant to be drawn at a specific time and is drawn at another time instead, the result can look abnormal even in a healthy person. This is why some hormone tests specify a required collection time.

#322 · Pre-Analytical Phase

Why does exercise before a blood draw sometimes affect results?

Answer

Exercise can temporarily raise levels of substances like creatine kinase, lactate, and certain white blood cell counts due to muscle activity and fluid shifts. It can also affect potassium and glucose levels. Patients are often asked to avoid strenuous activity before certain tests for this reason.

#323 · Pre-Analytical Phase

How can patient posture affect test results, especially protein-based tests?

Answer

When a person stands up, fluid shifts out of the blood vessels into surrounding tissue, which concentrates larger molecules like proteins and blood cells in the remaining blood. This is called postural or orthostatic change, and it can raise results for total protein, albumin, and hematocrit compared to lying down. Standardizing patient position before collection helps reduce this variability.

#324 · Pre-Analytical Phase

Why does prolonged tourniquet application count as a biologic or physiologic pre-analytical variable?

Answer

Leaving a tourniquet on for more than about one minute causes fluid to leak out of the blood vessel into the tissue, concentrating cells and large molecules in the blood left behind. This can falsely raise results like potassium, protein, and hematocrit. Guidelines recommend releasing the tourniquet as soon as blood flow starts or within one minute.

#325 · Pre-Analytical Phase

How can diet or fasting status affect lab results?

Answer

Eating a meal, especially one high in fat, can raise triglycerides and cause lipemia that interferes with testing. Glucose levels also rise temporarily after eating. Many tests, such as fasting glucose and lipid panels, require the patient to fast for a set number of hours beforehand.

#326 · Pre-Analytical Phase

How does patient age act as a biologic variable in lab testing?

Answer

Normal reference ranges often differ between infants, children, and adults because organ function and enzyme levels change throughout life. For example, alkaline phosphatase is naturally higher in growing children due to bone growth. Age appropriate reference ranges must be used to interpret results correctly.

#327 · Pre-Analytical Phase

How can pregnancy act as a biologic variable affecting lab values?

Answer

Pregnancy causes normal physiologic changes such as increased blood volume, which can dilute some substances, and hormonal shifts that raise others, like certain thyroid binding proteins. These changes are expected and not signs of disease. Reference ranges for pregnant patients often differ from the general population.

#328 · Pre-Analytical Phase

How can altitude affect certain blood test results?

Answer

People living at high altitude have naturally lower oxygen availability in the air, so their bodies produce more red blood cells to compensate. This means hemoglobin and hematocrit results are often naturally higher in high altitude populations. This must be considered when interpreting whether a result is truly abnormal.

#329 · Pre-Analytical Phase

How can stress affect lab test results as a biologic variable?

Answer

Physical or emotional stress can trigger the release of hormones like cortisol and adrenaline, which in turn can raise glucose, white blood cell count, and certain other markers. This response is a normal body reaction, not a lab error. It is one reason patients are often asked to relax before a blood draw.

#330 · Pre-Analytical Phase

How do medications count as a variable affecting pre-analytical results?

Answer

Many drugs can change lab values directly or interfere with the testing method itself, separate from any collection error. For example, aspirin can affect platelet function tests, and biotin supplements can interfere with certain immunoassays. Lab assistants should note relevant medication use when it is provided, since it can explain an unusual result.

#331 · Pre-Analytical Phase

Why does patient sex act as a biologic variable in reference ranges?

Answer

Biological differences between males and females, such as muscle mass and hormone levels, cause some substances like creatinine and hemoglobin to have different normal ranges. Using the wrong sex-based reference range could make a normal result look abnormal or vice versa. This is a normal physiological difference, not an error in testing.

#332 · Pre-Analytical Phase

What is the difference between a pre-analytical biologic variable and a pre-analytical handling error?

Answer

A biologic variable is a true, expected difference caused by the patient's own body, such as age, sex, or recent exercise. A handling error is a mistake made during collection or processing, such as prolonged tourniquet time, wrong tube, or delayed processing. Both can change results, but only handling errors can usually be prevented by following correct procedure.

#333 · Pre-Analytical Phase

Why is smoking sometimes noted as a variable that can affect certain test results?

Answer

Smoking can raise white blood cell counts, carbon monoxide levels in the blood, and certain lipid values compared to a non-smoker. These changes reflect a real physiological response to nicotine and other chemicals in smoke. Some tests may ask about smoking history for accurate interpretation.

#334 · Pre-Analytical Phase

Why is it important to check a Gram stained slide for the correct decolorization time using a known control organism?

Answer

Using a control slide with a known gram positive and gram negative organism lets the lab assistant confirm the reagents and technique are working correctly before trusting patient results. If the control does not stain as expected, the whole batch of slides may be inaccurate. This quality check helps catch reagent or timing problems early.

#335 · Pre-Analytical Phase

Why must slides for microbiology smears, like Gram stains, be heat or methanol fixed before staining?

Answer

Fixing kills most organisms and firmly attaches the specimen to the glass slide so it does not wash away during the staining and rinsing steps. Heat fixation is done by briefly passing the slide through a flame, while methanol fixation uses a chemical instead. Without fixing, the sample could be lost or the technician could be exposed to live organisms.

#336 · Analytical Phase

What is point-of-care testing (POCT)?

Answer

POCT means running a lab test right at the patient bedside or clinic instead of sending the sample to the main lab. Examples include glucose meters, urine dipsticks, and rapid antigen tests. An MLA may only perform these tests if delegated by an MLT or supervisor and must follow the site's approved procedure.

#337 · Analytical Phase

Before using a glucose meter on a patient, what must the MLA check?

Answer

Check that the meter has passed its daily quality control (QC) run with liquid controls in the expected range. Confirm the test strips are not expired and match the meter's code or lot number. If QC is out of range, the MLA must not use the meter and should report it to the supervising MLT right away.

#338 · Analytical Phase

Why do glucose meter strips need a code or calibration match?

Answer

Each box of strips is made in a batch that can react slightly differently to glucose in the blood. The meter needs to be told which batch it is reading so it calculates the right number. Using the wrong code can give a falsely high or low glucose result, which could lead to the wrong treatment.

#339 · Analytical Phase

What sample type is normally used for a bedside glucose test?

Answer

A small drop of capillary blood from a fingerstick is used, unless the site protocol calls for a different site like a heel stick on an infant. The first drop is often wiped away because it can contain tissue fluid that skews the result. Enough blood must fully cover the strip's test area or the meter will give an error.

#340 · Analytical Phase

What is an ECG (electrocardiogram)?

Answer

An ECG is a recording of the heart's electrical activity picked up through electrodes stuck to the skin. It shows the rhythm and timing of each heartbeat as a wave pattern on paper or a screen. MLAs may help attach leads and run the machine only if this task has been delegated and trained on.

#341 · Analytical Phase

What is a Holter monitor used for?

Answer

A Holter monitor is a small portable ECG device the patient wears for 24 to 48 hours to record heart activity during normal daily life. It catches irregular heartbeats that might not show up during a short in-office ECG. The MLA's role, if delegated, is usually limited to applying electrodes and instructing the patient on care of the device.

#342 · Analytical Phase

Why is skin prep important before placing ECG electrodes?

Answer

Oil, lotion, sweat, and dead skin cells block good electrical contact between the skin and the electrode. Poor contact causes a noisy, unreadable tracing full of artifact. The site is usually wiped with alcohol and lightly abraded before the electrode is applied.

#343 · Analytical Phase

What does a urine chemistry dipstick screen for?

Answer

A urine dipstick has small pads that change color to screen for things like glucose, protein, blood, ketones, leukocytes, nitrites, pH, and specific gravity. It gives a fast rough answer, not a precise lab number. Abnormal results are usually confirmed or followed up with formal lab testing.

#344 · Analytical Phase

What is a common timing rule when reading a urine dipstick?

Answer

Each pad on the strip must be read at its own specific time, often somewhere between 30 and 120 seconds after dipping. Reading too early or too late can give a false color reaction. Many labs now use an automated dipstick reader to keep timing consistent.

#345 · Analytical Phase

How should a urine specimen be handled before dipstick testing?

Answer

The urine should be well mixed and at room temperature, and testing should happen soon after collection. Urine left sitting out too long can change pH, lose glucose or ketones, and grow bacteria that change the result. If a delay is unavoidable, the sample should be refrigerated per the site's protocol.

#346 · Analytical Phase

How does a urine pregnancy test work in simple terms?

Answer

The test strip detects human chorionic gonadotropin (HCG), a hormone that rises quickly in early pregnancy. Urine flows across the strip and reacts with antibodies that produce a colored line if HCG is present. A control line must always appear or the test is invalid and must be repeated.

#347 · Analytical Phase

What can cause a false negative urine pregnancy test?

Answer

Testing too early in the pregnancy before HCG has built up enough can miss a true pregnancy. Very dilute urine, such as first thing after drinking a lot of fluids, can also lower the HCG concentration below the detection limit. First morning urine is usually preferred because it is the most concentrated.

#348 · Analytical Phase

What is a COVID-19 rapid antigen test detecting?

Answer

It detects viral proteins from the surface of the SARS-CoV-2 virus, not the virus's genetic material like a PCR test does. A nasal or nasopharyngeal swab sample is mixed with a reagent and applied to a test strip. Results usually appear within 15 to 30 minutes, and both a control and test line must show for a valid result.

#349 · Analytical Phase

Why might a rapid antigen test miss an early infection?

Answer

Antigen tests need a certain amount of virus present to trigger a positive line, and viral levels can be too low in the first day or two of infection. This is why a single negative antigen test does not fully rule out infection. Repeat testing or a more sensitive PCR test may be recommended if symptoms continue.

#350 · Analytical Phase

What does 'reading past the result window' mean and why does it matter?

Answer

Every rapid test lists a time range, such as 15 to 30 minutes, during which the result is considered accurate. Reading before that window can miss a faint developing line, and reading after it can show a false line caused by the reagent drying out. The MLA must always follow the manufacturer's exact timing.

#351 · Analytical Phase

What is a routine stain used for in the lab, such as Wright or Wright-Giemsa stain?

Answer

This stain is used on blood smears so that different blood cells can be told apart under the microscope. It colors the nucleus and cytoplasm of white blood cells differently, which helps identify cell types and shapes. An MLA may prepare and stain slides but the actual differential count and interpretation is done by an MLT.

#352 · Analytical Phase

What does ESR stand for and what does it screen for?

Answer

ESR stands for erythrocyte sedimentation rate, a simple test that measures how fast red blood cells settle to the bottom of a tube in one hour. A faster than normal rate can be a sign of inflammation somewhere in the body. It is a nonspecific screening test, meaning it shows something may be wrong but not exactly what.

#353 · Analytical Phase

What can affect an ESR result if the tube is handled incorrectly?

Answer

The tube must sit perfectly upright and level for the full hour, since tilting speeds up how fast the cells settle and gives a falsely high result. Clots, bubbles, or an underfilled tube can also throw off the reading. Room temperature should stay steady because heat can speed up the settling process too.

#354 · Analytical Phase

What is meant by 'cellular versus non-cellular components' in lab testing?

Answer

Cellular components are the actual cells, like red cells, white cells, and platelets, that can be counted, stained, and examined for shape. Non-cellular components are the liquid parts, such as plasma or serum, that carry chemicals, hormones, and proteins. Different tests and techniques are used depending on whether the lab is looking at cells or the fluid around them.

#355 · Analytical Phase

What should an MLA check when verifying a stained microscopic slide before it goes for review?

Answer

Check that the smear has a proper thin feathered edge, is evenly spread with no clumping, and is fully dried before staining. Look that the stain color is correct, not too dark or too pale, and that the slide is clean and labeled correctly. If anything looks wrong, the MLA should flag it and ask the MLT rather than deciding on their own.

#356 · Analytical Phase

When should an MLA request help from an MLT during microscopic prep work?

Answer

Whenever a slide looks abnormal, understained, overstained, or the smear technique failed, the MLA should stop and ask for MLT support rather than guessing. Any result that seems clinically unusual or urgent should also be escalated right away. MLAs are trained to prepare and screen basics, not to make diagnostic interpretations.

#357 · Analytical Phase

What is plating a specimen in microbiology?

Answer

Plating means spreading a specimen, like urine or a swab sample, onto a culture plate containing nutrient agar so any bacteria present can grow into visible colonies. Different agar types support or suppress different bacteria to help sort out what is growing. An MLA may plate specimens only when this task has been delegated and the technique has been properly trained.

#358 · Analytical Phase

What does 're-plating' a specimen usually mean?

Answer

Re-plating means repeating the plating step, often because the first plate was contaminated, had too little growth, or the streak technique did not properly isolate single colonies. It can also be requested if the wrong agar type was used the first time. This is done under supervision and following the lab's exact protocol for streaking technique.

#359 · Analytical Phase

What is the streak plate technique used for?

Answer

It is a method of spreading a small amount of specimen across an agar plate in a specific pattern, usually in three or four sections, to thin out the bacteria. This spacing lets individual bacteria grow into separate, isolated colonies instead of one solid smear of growth. Isolated colonies are needed so the lab can identify and test one type of bacteria at a time.

#360 · Analytical Phase

What is a basic testing principle behind most point-of-care devices?

Answer

Most POCT devices use a chemical reaction, like an enzyme reacting with glucose or an antibody binding to a target protein, that produces a measurable color change or electrical signal. The device reads that change and converts it into a number or line result. Understanding this helps the MLA recognize when a result looks physically impossible and may signal a device or sample problem.

#361 · Analytical Phase

What is the first troubleshooting step if a point-of-care device gives an error message?

Answer

Check the simple things first, such as whether enough sample was applied, whether the strip or cartridge is expired, and whether the device battery is charged. Confirm the device passed its most recent quality control check. If the error continues after these checks, the MLA should stop and report it to the supervising MLT rather than keep repeating the test.

#362 · Analytical Phase

Why is running liquid quality control (QC) material important before patient testing?

Answer

QC material has a known expected result, so running it confirms the device and reagents are working correctly before a real patient sample is tested. If the QC result falls outside its expected range, the device cannot be trusted and must not be used on patients. All QC results should be documented, not just glanced at and dismissed.

#363 · Analytical Phase

What is meant by sensitivity and specificity in simple terms for a screening test?

Answer

Sensitivity is how good a test is at correctly catching people who truly have the condition, so a highly sensitive test rarely misses a true case. Specificity is how good a test is at correctly clearing people who truly do not have the condition, so it rarely gives a false alarm. Screening tests like rapid antigen or dipstick tests often trade off some specificity for higher sensitivity.

#364 · Analytical Phase

What is a control line on a lateral flow test strip, like a pregnancy or antigen test?

Answer

The control line is a built-in check showing that the test strip worked properly and that fluid flowed across it as expected. If the control line does not appear, the whole test is invalid no matter what the test line shows. An invalid test must be repeated with a new strip, not reported as a result.

#365 · Analytical Phase

Why must an MLA always work under MLT or supervisor oversight for analytical tasks?

Answer

MLAs are trained to help with sample handling and basic testing steps, but the scope of their role does not include independent interpretation of clinical results. A supervising MLT or other qualified professional reviews and signs off on results that affect patient care. This layered check helps catch errors before they reach the doctor or patient.

#366 · Analytical Phase

What documentation should go along with every point-of-care test result?

Answer

Record the patient identifier, date and time, test result, lot number of the strip or cartridge, and the operator's initials. Any QC run that day should also be logged. Good documentation lets someone trace back and check for a problem if a result is later questioned.

#367 · Analytical Phase

What is hemolysis and why does it matter for POCT and other analytical testing?

Answer

Hemolysis is when red blood cells break open and release their contents, like hemoglobin and potassium, into the surrounding fluid. This can happen from squeezing a fingerstick site too hard or from a rough sample draw. A hemolyzed sample can give falsely high or low results on many tests and may need to be rejected and redrawn.

#368 · Analytical Phase

What is the purpose of a lockbox or two-step verification step for critical POCT results?

Answer

Some devices require a second check, like scanning an operator ID or entering a supervisor code, before a critical result can be released. This ensures a trained and authorized person confirms the result before it reaches the patient's chart. It adds a safety layer for results that could quickly affect patient treatment, like a very high or low glucose.

#369 · Analytical Phase

Why should test strips and cartridges be stored properly, such as away from heat, light, or moisture?

Answer

The chemicals on strips and cartridges can break down if exposed to heat, humidity, or light, which changes how they react during testing. A damaged strip can give an inaccurate result even if it is not visibly expired. Containers should be kept sealed and stored exactly as the manufacturer's insert instructs.

#370 · Analytical Phase

If a point-of-care result does not match how the patient looks clinically, what should the MLA do?

Answer

The MLA should not simply report the number and move on if it seems inconsistent with the patient's obvious condition, such as a normal glucose reading in a clearly symptomatic patient. The result and situation should be reported to the supervising MLT or appropriate professional right away. Retesting or sending a sample to the main lab may be needed to confirm the true result.

#371 · Post-Analytical Phase

What is the post-analytical phase of lab testing?

Answer

It is everything that happens after a test has actually been run on the sample. This includes recording the result, reviewing it for accuracy, and getting it to the right place for verification. For an MLA, this phase is mostly about careful documentation, not interpretation.

#372 · Post-Analytical Phase

Can an MLA interpret or release a final patient result on their own?

Answer

No. An MLA can record and pass along results, but a Medical Laboratory Technologist (MLT) must review and validate the result before it is considered final. This split protects patients because the MLA's scope of practice does not include clinical interpretation.

#373 · Post-Analytical Phase

Why must results be recorded only after MLT validation?

Answer

Recording a result before it is validated risks putting an unchecked or possibly wrong number into the patient chart. Once a result is validated, it can be trusted as accurate and ready for clinical use. Waiting for validation keeps a clear line between preliminary data and an official result.

#374 · Post-Analytical Phase

What does LIS stand for?

Answer

LIS stands for Laboratory Information System. It is the computer system labs use to track samples, store results, and send information to doctors and hospital charts. MLAs use the LIS constantly to log samples in, check statuses, and enter recorded results.

#375 · Post-Analytical Phase

What should an MLA check before entering a result into the LIS?

Answer

They should check that the result is linked to the correct patient, the correct test, and the correct sample ID. They should also make sure the value looks complete and was actually authorized by the MLT. A small mismatch here can send the wrong result to the wrong patient's chart.

#376 · Post-Analytical Phase

What are the three things an MLA verifies about a result before passing it along?

Answer

Accuracy, completeness, and clarity. Accuracy means the number matches what was reported by the instrument or MLT. Completeness means no fields are missing, and clarity means the entry is legible and free of confusing abbreviations.

#377 · Post-Analytical Phase

What counts as an incomplete result entry?

Answer

An incomplete entry is missing a required piece of information, such as the units of measurement, the specimen type, or the time the test was performed. It can also mean a field was left blank when it should have data in it. Incomplete entries should be flagged and corrected before moving forward, not guessed at.

#378 · Post-Analytical Phase

Why does clarity matter in a recorded result?

Answer

A result that is smudged, cut off, or written with unclear handwriting or shorthand can be misread by whoever reviews it next. Misreading a number like a 1 for a 7 can lead to a serious clinical mistake. Clear, legible recording protects the patient down the line.

#379 · Post-Analytical Phase

What is the MLA's role when a result comes back looking abnormal or critical?

Answer

The MLA records what they see and immediately brings it to the attention of the MLT or supervisor rather than deciding what it means. They do not delay passing along an unusual result. Fast, accurate reporting lets the qualified technologist act on it right away.

#380 · Post-Analytical Phase

Define 'delegated task' in the context of recording results.

Answer

A delegated task is one that a supervising MLT or lab professional has specifically assigned to the MLA, such as data entry or transcription. It does not include making judgment calls about the result itself. Staying inside the delegated task keeps the MLA working within their approved scope.

#381 · Post-Analytical Phase

What should an MLA do if they notice a typo in a result they already entered?

Answer

They should correct it right away using the lab's approved correction procedure, which usually involves flagging the original entry rather than deleting it. Silently overwriting a result destroys the audit trail. Reporting the correction to the MLT keeps everyone informed.

#382 · Post-Analytical Phase

What is an audit trail?

Answer

An audit trail is the record of every change made to a result, showing who entered it, who changed it, and when. It exists so that any error can be traced back to its source. The LIS usually keeps this trail automatically for every entry and edit.

#383 · Post-Analytical Phase

Why should an MLA never delete an entry instead of correcting it?

Answer

Deleting removes proof that the value ever existed, which breaks the audit trail and can hide an error instead of documenting it. Labs require a visible correction process so mistakes stay transparent. This protects both the patient and the lab if the record is ever reviewed.

#384 · Post-Analytical Phase

What does 'providing results to the MLT for verification' mean in practice?

Answer

It means the MLA hands off the raw or recorded result, along with any relevant sample notes, to the MLT so they can check it before it becomes official. This step is not optional, even if the result looks normal. Every result passes through this checkpoint.

#385 · Post-Analytical Phase

What is a 'rare extenuating circumstance' scenario for blood product issuing?

Answer

This refers to an unusual emergency situation, such as a life-threatening bleed, where local policy allows an MLA to assist with issuing blood products under strict supervision. It is not a normal daily task and only applies when clearly permitted by facility protocol. Even then, an MLT or physician oversees the process.

#386 · Post-Analytical Phase

Why is blood product issuing normally outside the MLA scope of practice?

Answer

Issuing blood products carries a high risk of serious harm if the wrong unit goes to the wrong patient, so it requires specialized cross-matching knowledge that MLTs are trained for. This makes it a task reserved for qualified technologists in almost all situations. The rare exception only applies under documented emergency protocols.

#387 · Post-Analytical Phase

What double-check is essential before any blood product leaves the lab, even in an emergency?

Answer

A second qualified person must confirm the patient identity, blood type compatibility, and unit number match exactly. This two-person check happens no matter how urgent the situation is. Skipping it risks a fatal transfusion reaction.

#388 · Post-Analytical Phase

What is a critical value?

Answer

A critical value is a result so far outside the normal range that it could indicate a life-threatening condition needing immediate action. Labs have strict rules for how fast these must be reported and to whom. An MLA who spots one flags it immediately rather than waiting for a routine review cycle.

#389 · Post-Analytical Phase

How should an MLA handle a discrepancy between a written result and what is shown on the analyzer screen?

Answer

They should stop and report the mismatch to the MLT instead of picking one value to enter. Entering either number without resolving the discrepancy could put a wrong result in the patient's file. The MLT decides how to resolve it, often by rechecking the sample.

#390 · Post-Analytical Phase

What is the purpose of a results log or worksheet in the post-analytical phase?

Answer

It gives a written or digital backup record of results as they come off the instrument, before they are entered into the LIS. This lets staff cross-check the LIS entry against the original reading. It is an extra layer of protection against transcription errors.

#391 · Post-Analytical Phase

Why should an MLA avoid using unapproved abbreviations when recording results?

Answer

Unapproved abbreviations can be misread by other staff or mean something different in another department, leading to confusion or errors. Labs keep an approved abbreviation list for this reason. Sticking to that list keeps records clear for anyone who reads them later.

#392 · Post-Analytical Phase

What should an MLA do if the LIS is down and results cannot be entered electronically?

Answer

They follow the facility's downtime procedure, which usually means recording results on paper forms in a specific, approved format. These paper records are then entered into the LIS once it is back online, following the same verification steps. Nothing gets skipped just because the system is temporarily unavailable.

#393 · Post-Analytical Phase

Why is timeliness important in the post-analytical phase, even for an MLA's limited tasks?

Answer

A delayed result can hold up a doctor's decision about patient care, especially for urgent tests. MLAs are expected to record and pass along results promptly once instructed to do so. Being fast should never come at the cost of accuracy, but unnecessary delay is also a problem.

#394 · Post-Analytical Phase

What is the difference between a preliminary result and a validated result?

Answer

A preliminary result is what the instrument or an early check produces, and it has not yet been reviewed by an MLT. A validated result has been checked and approved by the MLT and is ready to go to the patient's chart. An MLA should never treat a preliminary number as final.

#395 · Post-Analytical Phase

Overall, what is the guiding principle for an MLA during the post-analytical phase?

Answer

Stay inside the delegated task of accurate, complete, and clear recording, and always route anything uncertain to the MLT rather than deciding it yourself. This keeps the patient safe and keeps the MLA working within their proper scope. When in doubt, ask rather than assume.

#396 · Quality and Resource Management

What is internal quality control (QC) in a lab?

Answer

Internal QC means running control samples with known expected values alongside patient samples on the same test run. If the control result falls within the accepted range, the run is considered valid. If it falls outside that range, the lab must troubleshoot before releasing patient results.

#397 · Quality and Resource Management

What is external quality control (proficiency testing)?

Answer

External QC involves testing unknown samples sent by an outside agency and reporting the results back to that agency for comparison against other labs. It checks how a lab performs compared to peers using the same method, not just against its own internal controls. Poor performance can trigger a corrective action plan.

#398 · Quality and Resource Management

Why must QC data always be documented, not just reviewed?

Answer

Written or electronic QC records create a permanent trail proving that testing was monitored correctly on a given day. If a result is ever questioned later, the documentation shows whether controls were in range at the time. Accreditation bodies require these records to be kept and available for audit.

#399 · Quality and Resource Management

What should an MLA do if a QC result is out of range?

Answer

The MLA should not release patient results from that run. Instead they should notify the supervisor or technologist, recheck the control, and look for causes such as expired reagents, a calibration problem, or operator error. The issue must be resolved and documented before testing continues.

#400 · Quality and Resource Management

Define quality control (QC).

Answer

Quality control is the set of specific tests and checks done to confirm that an individual test run or instrument is working correctly at the time of testing. It focuses on the technical accuracy of results in the moment, using controls with known values.

#401 · Quality and Resource Management

Define quality assurance (QA).

Answer

Quality assurance is the broader system of policies, procedures, and monitoring that make sure the whole testing process, from specimen collection to result reporting, is reliable over time. QA looks at the big picture, while QC checks one specific run or test.

#402 · Quality and Resource Management

How do QC and QA differ in scope?

Answer

QC is narrow and immediate, it tells you if today's test run is working right. QA is broad and ongoing, it covers training, specimen handling, equipment maintenance, and error tracking across the whole lab. QC is one tool inside the larger QA system.

#403 · Quality and Resource Management

What is calibration in the lab setting?

Answer

Calibration is the process of adjusting an instrument using materials with known, certified values so its readings match the true value. It is usually done on a set schedule or after certain events like a major repair. Without regular calibration, an instrument can drift and give inaccurate results even if it seems to be working.

#404 · Quality and Resource Management

When does an instrument usually need recalibration outside of the routine schedule?

Answer

Recalibration is needed after major maintenance or repairs, after a lot has changed, after QC results start trending or falling outside range, or after the instrument has been moved. Recalibrating restores confidence that results are accurate before patient testing resumes.

#405 · Quality and Resource Management

How does an MLA verify the quality of a new reagent lot before use?

Answer

The MLA runs QC controls using the new reagent lot and compares the results to the expected ranges and to the old lot's performance. Only if the new lot passes should it be put into routine use. This step catches manufacturing problems or shipping damage before patient samples are affected.

#406 · Quality and Resource Management

What signs indicate a culture media plate has gone bad?

Answer

Signs include visible contamination, cracking or drying of the agar, discoloration, an unusual odor, or the expiry date having passed. Media should also be checked with known control organisms periodically to confirm it still supports expected growth.

#407 · Quality and Resource Management

Why is media quality control important in microbiology?

Answer

Media that has lost its nutrients or become contaminated may fail to grow an organism that is actually present, leading to a false negative result. Testing media with known positive and negative control organisms confirms the media still performs correctly before patient specimens are plated.

#408 · Quality and Resource Management

What is continuous quality improvement (CQI)?

Answer

CQI is an ongoing process of looking for small ways to make lab operations better, safer, and more efficient rather than just fixing problems as they occur. It relies on collecting data, spotting patterns, and making gradual planned changes. The goal is steady improvement over time, not a one time fix.

#409 · Quality and Resource Management

Give an example of a CQI activity in a lab.

Answer

Tracking turnaround times for stat tests each month and looking for trends is a CQI activity. If times are slowly getting longer, the team can investigate causes like staffing or equipment before it becomes a serious problem. This proactive review is different from just reacting to a single complaint.

#410 · Quality and Resource Management

What is an occurrence report (incident report) used for?

Answer

An occurrence report documents any error, accident, or unusual event that happens during testing or patient care, such as a mislabeled sample or a needlestick. It records what happened, when, and who was involved so the issue can be reviewed and corrected. These reports are part of the lab's risk management system.

#411 · Quality and Resource Management

What is risk management in a laboratory context?

Answer

Risk management is the process of identifying things that could go wrong, such as specimen mix ups or equipment failure, and putting steps in place to prevent or reduce them. It also includes reviewing errors after they happen to stop them from repeating. The goal is fewer mistakes and safer patient care.

#412 · Quality and Resource Management

Why should an MLA report a near miss even if no harm occurred?

Answer

A near miss shows a weakness in the process that could cause real harm next time if left uncorrected. Reporting it lets the lab fix the root cause before a patient is actually affected. This is a key part of a proactive risk management culture.

#413 · Quality and Resource Management

What is a root cause analysis?

Answer

Root cause analysis is a structured way of investigating an error to find the true underlying reason it happened, rather than just blaming the last person who touched it. It often involves asking why repeatedly until the real system problem is found. Fixing the root cause prevents the same error from recurring.

#414 · Quality and Resource Management

What is meant by resource management in the lab?

Answer

Resource management means making sure staff, supplies, equipment, and time are used efficiently so the lab can meet testing demands. It includes scheduling staff appropriately and making sure enough trained people are available for the workload. Poor resource management can lead to delays or errors.

#415 · Quality and Resource Management

What is time management important for an MLA's daily duties?

Answer

MLAs often juggle specimen collection, processing, and clerical tasks under strict turnaround time expectations, especially for stat orders. Prioritizing urgent samples first and organizing routine work around them keeps the lab running smoothly. Poor time management can delay critical patient results.

#416 · Quality and Resource Management

What does inventory management involve in the lab?

Answer

Inventory management means tracking supply levels, checking expiry dates, rotating stock using first in first out order, and reordering before items run out. Proper inventory management prevents testing delays caused by missing supplies and reduces waste from expired reagents. It is a routine responsibility often assigned to MLAs.

#417 · Quality and Resource Management

What does first in, first out (FIFO) mean for supply storage?

Answer

FIFO means using the oldest stock first so that reagents and supplies are used before they expire. New shipments are placed behind existing stock rather than in front of it. This simple rotation habit reduces the amount of expired material that has to be thrown away.

#418 · Quality and Resource Management

Why should reagents and supplies be checked against a par level?

Answer

A par level is the minimum amount of a supply that should always be on hand to avoid running out. Checking stock against par levels regularly tells staff when it is time to reorder. This prevents test delays caused by suddenly discovering an item is out of stock.

#419 · Quality and Resource Management

What is change management in a laboratory setting?

Answer

Change management is the planned process labs use when introducing something new, like a new instrument, test method, or software system. It typically includes validation, staff training, and a trial period before the change fully replaces the old process. This structured approach reduces errors caused by an unplanned or rushed switch.

#420 · Quality and Resource Management

Why is staff training a required step before implementing a lab change?

Answer

Even a good new method or instrument can cause errors if staff do not understand how to use it correctly. Training ensures everyone can perform the new process safely and consistently before it goes live. Skipping training increases the risk of mistakes during the transition period.

#421 · Quality and Resource Management

What is a Levey-Jennings chart used for?

Answer

A Levey-Jennings chart is a graph that plots QC results over time against the expected mean and standard deviation limits. It helps staff visually spot trends or shifts in control values before they cause a real problem. It is a common tool for monitoring QC over days or weeks.

#422 · Quality and Resource Management

What does a QC value showing a gradual trend upward or downward suggest?

Answer

A gradual trend, even if each value is still technically within range, can suggest a slow problem developing such as reagent deterioration or instrument drift. It is different from a sudden shift, which often points to a specific one time event like a new lot or calibration change. Either pattern should prompt investigation before it leads to inaccurate patient results.

#423 · Quality and Resource Management

What is a corrective action plan?

Answer

A corrective action plan is a documented set of steps taken to fix the cause of a problem, such as a failed proficiency test or repeated QC failures. It states what went wrong, what will be done to fix it, and how the fix will be checked for success. Accrediting bodies often require these plans on file.

#424 · Quality and Resource Management

Why do accreditation standards require ongoing competency assessment of staff?

Answer

Competency assessment confirms that staff continue to perform tests correctly over time, not just when they were first trained. It might include direct observation, review of QC records, or blind testing with unknown samples. This protects patients by catching skill gaps before they cause errors.

#425 · Quality and Resource Management

What is the purpose of a standard operating procedure (SOP) manual?

Answer

An SOP manual gives written, step by step instructions for every test and process performed in the lab so results are consistent no matter who performs the work. It is a key reference during training, troubleshooting, and audits. Keeping SOPs updated is part of the lab's overall quality system.

#426 · Quality and Resource Management

What is a control chart limit, often called a Westgard rule, used for?

Answer

Westgard rules are a set of statistical criteria applied to QC data to decide whether a run should be accepted or rejected. They look at patterns like one control being far outside range or several controls trending the same direction. Using these rules helps labs catch both random and systematic errors.

#427 · Quality and Resource Management

How does documenting equipment maintenance support quality management?

Answer

Maintenance logs show that instruments were cleaned, serviced, and checked on schedule, which supports confidence in the results they produce. If a problem arises later, the log helps determine whether missed maintenance could be the cause. This documentation is also reviewed during accreditation inspections.

#428 · Quality and Resource Management

Why is turnaround time (TAT) tracked as a quality indicator?

Answer

Turnaround time measures how long it takes from specimen collection to result reporting, and slow TAT can delay patient care decisions. Tracking it over time helps the lab spot bottlenecks in staffing, equipment, or workflow. It is a common metric reviewed as part of continuous quality improvement.

#429 · Quality and Resource Management

What role does an MLA typically play in the lab's quality system?

Answer

MLAs often run and document daily QC checks, monitor supply and reagent expiry dates, report occurrences or errors, and follow SOPs exactly as written. While they are not usually the ones designing the quality program, their consistent day to day compliance is essential to it working. Attention to detail at this level catches many problems early.

#430 · Quality and Resource Management

Why must expired reagents or controls never be used, even if supply is low?

Answer

Once a reagent or control passes its expiry date, its chemical stability is no longer guaranteed, so results could be inaccurate even if it looks and behaves normally. Using expired materials to save time or stock risks patient safety and can invalidate the whole test run. Running low on stock should trigger reordering, not use of expired material.

#431 · Communication and Collaboration

What is active listening?

Answer

Active listening means giving full attention to the speaker, not just waiting for your turn to talk. It includes making eye contact, nodding, and repeating back what you heard to confirm understanding. This helps patients feel heard and reduces mistakes caused by mishearing instructions.

#432 · Communication and Collaboration

Why should a lab assistant avoid medical jargon with patients?

Answer

Most patients are not trained in medical terms and jargon can confuse or scare them. Using plain language helps the patient understand what is happening and why. It also builds trust because the patient feels respected rather than talked down to.

#433 · Communication and Collaboration

What is the teach-back method?

Answer

The teach-back method asks the patient to repeat instructions back in their own words. This confirms they actually understood, rather than just saying yes. It is especially useful for fasting instructions or specimen collection steps done at home.

#434 · Communication and Collaboration

Define nonverbal communication in patient care.

Answer

Nonverbal communication includes facial expressions, posture, tone of voice, and gestures. Patients often pick up on these cues even more than spoken words. A calm tone and relaxed posture can ease patient anxiety during blood draws.

#435 · Communication and Collaboration

What should you do if a patient speaks limited English?

Answer

Use a qualified medical interpreter or translation service rather than a family member when possible. Family members may not know medical terms correctly or may filter information. Facilities usually have phone interpreter lines available for many languages.

#436 · Communication and Collaboration

Why is using a family member as an interpreter risky?

Answer

Family members are not trained in medical terminology and may translate incorrectly or leave out details. They may also feel uncomfortable relaying sensitive information, which can lead to the patient hiding facts. A professional interpreter protects accuracy and patient privacy.

#437 · Communication and Collaboration

What is empathy in patient interaction?

Answer

Empathy means recognizing and understanding what a patient is feeling, such as fear of needles. Showing empathy might sound like acknowledging the patient's nervousness before starting a draw. It is different from sympathy, which is feeling sorry for someone rather than understanding them.

#438 · Communication and Collaboration

How should a lab assistant greet a new patient?

Answer

Introduce yourself by name and role, confirm the patient's identity using two identifiers, and explain what procedure you will perform. This sets expectations and reduces anxiety. A friendly, calm introduction also builds immediate trust.

#439 · Communication and Collaboration

What is cultural safety?

Answer

Cultural safety means creating an environment where patients feel respected regardless of their cultural background, and free from judgment or discrimination. It goes beyond just being aware of differences, it focuses on the patient's own experience of care. The patient, not the provider, decides if care felt culturally safe.

#440 · Communication and Collaboration

What is cultural humility?

Answer

Cultural humility is an ongoing process of self reflection about your own biases and assumptions, rather than claiming to be an expert on another culture. It means staying open to learning from each patient as an individual. This lifelong approach helps avoid stereotyping.

#441 · Communication and Collaboration

How does cultural humility differ from cultural competence?

Answer

Cultural competence often implies you can master knowledge about a culture and be done learning. Cultural humility treats learning as continuous and recognizes you can never fully know another person's lived experience. It emphasizes humility and respect over checklist knowledge.

#442 · Communication and Collaboration

Give an example of inclusive behaviour in the lab.

Answer

Inclusive behaviour includes asking patients their preferred name and pronouns rather than assuming. It also means adapting communication for patients with disabilities, such as facing a deaf patient so they can read lips or using written notes. These small actions make all patients feel welcome.

#443 · Communication and Collaboration

What should you do if a patient refuses a procedure?

Answer

Calmly ask why they are hesitant and address their concerns with clear information. Respect their right to refuse and document the refusal according to policy. Notify the supervisor or ordering provider so follow up can happen.

#444 · Communication and Collaboration

What is a conflict management technique for a frustrated patient?

Answer

Stay calm, lower your voice, and listen without interrupting to let the patient feel heard. Acknowledge their frustration before explaining next steps or solutions. Avoid arguing back, since a calm response often de-escalates the situation faster than defending yourself.

#445 · Communication and Collaboration

What is de-escalation?

Answer

De-escalation is a set of techniques used to reduce tension in a heated situation before it becomes a bigger conflict. It includes speaking softly, giving the person space, and validating their feelings. The goal is to calm the situation, not to win an argument.

#446 · Communication and Collaboration

How should disagreements with a coworker be handled?

Answer

Address the issue directly and privately with the coworker first, using respectful language focused on the specific problem. If it cannot be resolved, involve a supervisor. Avoid discussing the conflict in front of patients or other staff.

#447 · Communication and Collaboration

Why is interprofessional collaboration important in health care?

Answer

Lab assistants work alongside nurses, phlebotomists, technologists, and physicians, all of whom depend on accurate and timely lab work. Good collaboration ensures specimens are collected, labelled, and delivered correctly. Poor communication between team members can lead to delays or patient safety errors.

#448 · Communication and Collaboration

What is SBAR and why is it used?

Answer

SBAR stands for Situation, Background, Assessment, Recommendation. It is a structured way to communicate important information clearly between health care team members. Using SBAR reduces confusion when reporting an urgent issue like a critical lab value.

#449 · Communication and Collaboration

What should you do when reporting a critical lab value?

Answer

Report it immediately to the ordering provider or nurse by phone, following facility policy for time limits. Read back the value to confirm accuracy with the receiver. Document who you spoke with, the time, and the value reported.

#450 · Communication and Collaboration

What is closed loop communication?

Answer

Closed loop communication means the receiver repeats back what they heard so the sender can confirm it is correct. This prevents errors from mishearing verbal orders or results. It is especially important for urgent or critical information.

#451 · Communication and Collaboration

How can body language affect a nervous patient during blood collection?

Answer

A tense posture or rushed movements can increase a patient's anxiety, sometimes triggering a vasovagal reaction. Moving slowly and speaking in a calm, steady voice helps the patient relax. Explaining each step before doing it also reduces surprise and fear.

#452 · Communication and Collaboration

What is the difference between hearing and listening?

Answer

Hearing is simply receiving sound, while listening involves actively processing and understanding the meaning behind the words. A lab assistant can hear a patient speak without truly listening to their concerns. Effective communication requires full listening, not just hearing.

#453 · Communication and Collaboration

How should you communicate with a pediatric patient?

Answer

Use simple, age appropriate language and involve the caregiver in the explanation. Distraction techniques, like a toy or a story, can help calm a child during a procedure. Speaking directly to the child, not just the parent, shows respect for the child as a person.

#454 · Communication and Collaboration

How should you adjust communication for an elderly patient with hearing loss?

Answer

Face the patient directly so they can see your lips and expressions, and speak clearly without shouting. Reduce background noise if possible and check that any hearing aids are in place. Writing key information down can also help if verbal communication is difficult.

#455 · Communication and Collaboration

What is a barrier to communication in health care?

Answer

Barriers include language differences, hearing or vision impairment, pain, fear, and cultural differences in expressing symptoms. Environmental barriers like noise or lack of privacy can also interfere. Recognizing these barriers helps the lab assistant adjust their approach.

#456 · Communication and Collaboration

What is an example of a cultural consideration in patient interaction?

Answer

Some cultures consider direct eye contact disrespectful, while others expect it as a sign of honesty. Some patients may prefer a same gender health care worker for personal procedures. Being aware of these differences without assuming every patient from a group feels the same way is key.

#457 · Communication and Collaboration

How should a lab assistant respond to an angry patient without becoming defensive?

Answer

Acknowledge the patient's feelings first, such as saying you understand the wait has been frustrating. Avoid taking the anger personally and focus on solving the problem. Staying calm models professionalism and often reduces the patient's anger.

#458 · Communication and Collaboration

What documentation habit supports good team communication?

Answer

Writing clear, accurate, and timely notes ensures the next team member has the information they need. Vague or delayed documentation can cause repeated tests or missed critical information. Good documentation is a form of communication across shifts and departments.

#459 · Communication and Collaboration

Why is confidentiality part of respectful communication?

Answer

Discussing patient information only with those who need it for care shows respect for the patient's privacy and builds trust. Talking about patients in public areas like elevators or break rooms can break confidentiality even without naming them. Respecting privacy is both an ethical and communication skill.

#460 · Communication and Collaboration

What is the goal of patient centered communication?

Answer

Patient centered communication focuses on the individual patient's needs, values, and preferences rather than a one size fits all approach. It involves asking questions and listening rather than just giving information. This approach improves patient satisfaction and cooperation with procedures.

#461 · Communication and Collaboration

How should you handle a language barrier when no interpreter is immediately available?

Answer

Use simple words, gestures, pictures, or translated written materials if available while arranging for a proper interpreter. Avoid rushing the procedure since misunderstanding can lead to errors. Document that language assistance was needed and how it was addressed.

#462 · Communication and Collaboration

What is an example of unconscious bias affecting patient care?

Answer

Unconscious bias might lead a worker to assume a patient's pain tolerance or health literacy based on appearance or accent. This can result in unequal care or communication that feels dismissive. Reflecting on personal assumptions is part of practicing cultural humility.

#463 · Communication and Collaboration

How should a lab assistant give instructions for a fasting blood test?

Answer

Explain clearly how many hours of fasting are needed and what counts as fasting, such as water being allowed but not coffee. Confirm the patient understands using teach-back. Provide written instructions when possible so the patient can reference them at home.

#464 · Communication and Collaboration

What role does tone of voice play in patient trust?

Answer

A warm, steady tone can reassure a nervous patient, while a flat or rushed tone can make them feel like a number rather than a person. Tone often communicates more than the actual words used. Patients frequently remember how they were made to feel during care.

#465 · Communication and Collaboration

How should you communicate a delay to a waiting patient?

Answer

Be honest about the reason for the delay and give a realistic estimate of wait time if possible. Apologize for the inconvenience without over explaining internal issues. Checking in periodically shows the patient they have not been forgotten.

#466 · Communication and Collaboration

What is an example of effective handoff communication between shifts?

Answer

A good handoff includes pending specimens, any patient concerns, and equipment issues that the next worker needs to know. Using a standard format, like SBAR, keeps handoffs consistent and complete. Missing handoff details can cause delays or repeated work.

#467 · Communication and Collaboration

Why is it important to ask open ended questions with patients?

Answer

Open ended questions, such as asking how the patient is feeling today, allow the patient to share more detail than a simple yes or no question. This can reveal concerns that closed questions might miss. It also shows genuine interest in the patient's wellbeing.

#468 · Communication and Collaboration

How can a lab assistant show respect for a patient's religious or spiritual needs?

Answer

This might include allowing time for prayer before a procedure or being sensitive to modesty requirements during collection. Asking the patient directly about any needs shows respect rather than assuming based on appearance. Small accommodations can greatly improve patient comfort and trust.

#469 · Communication and Collaboration

What should you do if you do not understand a coworker's instructions?

Answer

Ask for clarification right away rather than guessing, especially when it involves patient safety or specimen handling. It is better to ask a question that seems simple than to make an error from assumption. Clear communication protects both the patient and the team.

#470 · Communication and Collaboration

How does collaboration improve turnaround time for lab results?

Answer

When lab assistants, technologists, and clinical staff communicate clearly about priorities, urgent results reach providers faster. Poor collaboration can cause specimens to sit unprocessed or results to go unreported. Strong teamwork directly supports timely and accurate patient care.

#471 · Professional Practice

What does professional accountability mean for a Medical Laboratory Assistant?

Answer

It means owning the results of your own actions, both good and bad. If you make a mistake, like mislabeling a tube, you report it right away instead of hiding it. Accountability protects patients and keeps trust between you, your coworkers, and your employer.

#472 · Professional Practice

Define conflict of interest in a lab setting.

Answer

A conflict of interest happens when your personal interests could affect your professional judgment. For example, testing a family member's sample and skipping proper steps because you assume it is fine. The right move is to disclose the situation and let someone else handle the task.

#473 · Professional Practice

What is meant by professional presence?

Answer

Professional presence is how you carry yourself at work through appearance, tone of voice, and body language. It includes wearing your uniform and ID badge correctly, speaking respectfully, and staying calm under pressure. Patients often judge the quality of care by how professional the staff appears.

#474 · Professional Practice

What is a code of ethics?

Answer

A code of ethics is a written set of values and rules that guide how a professional should behave, such as honesty, respect, and fairness. CSMLS has its own code of ethics that MLAs are expected to follow. It is different from a code of conduct, which lists specific rules rather than broad values.

#475 · Professional Practice

How is a code of conduct different from a code of ethics?

Answer

A code of conduct lists specific, practical rules of behaviour, like dress code or attendance policy. A code of ethics describes the broader values behind those rules, like honesty and respect for patients. Both work together to guide professional behaviour in the lab.

#476 · Professional Practice

What does confidentiality mean for lab staff?

Answer

Confidentiality means keeping patient information private and only sharing it with people who need it for care. This includes not discussing patient results in public areas like elevators or cafeterias. Breaking confidentiality can lead to disciplinary action and loss of patient trust.

#477 · Professional Practice

An MLA overhears a coworker discussing a patient's HIV status in the break room. What should the MLA do?

Answer

The MLA should politely remind the coworker that this is confidential information and should not be discussed outside of care needs. If the behaviour continues, the MLA should report it to a supervisor. Protecting patient privacy is everyone's responsibility, not just the person who collected the sample.

#478 · Professional Practice

What is scope of practice?

Answer

Scope of practice is the set of tasks and responsibilities a healthcare worker is trained and legally allowed to perform. For an MLA, this usually includes specimen collection and basic processing, not test interpretation. Staying inside scope keeps patients safe and protects the worker legally.

#479 · Professional Practice

A patient asks the MLA to explain what their lab results mean. How should the MLA respond?

Answer

The MLA should politely explain that interpreting results is outside their scope of practice. They should direct the patient to their doctor or another qualified professional for that discussion. This protects the patient from getting incorrect or incomplete information.

#480 · Professional Practice

What are professional boundaries?

Answer

Professional boundaries are the limits that keep the relationship between a healthcare worker and a patient appropriate and respectful. This means not sharing personal opinions, not forming personal relationships with patients, and keeping interactions focused on care. Crossing boundaries can harm patient trust and the worker's credibility.

#481 · Professional Practice

Why is it important to report unsafe practices even if a senior coworker is involved?

Answer

Patient safety comes before workplace comfort or hierarchy. If a senior staff member skips a safety step, like not verifying patient identity, it still needs to be reported through the correct channel. Ignoring unsafe practices because of seniority can lead to repeated harm.

#482 · Professional Practice

What is the first step an MLA should take when they witness unethical behaviour by a coworker?

Answer

The first step is usually to document what was observed, including date, time, and details. Then the MLA should report it following the workplace's official reporting process, often starting with a direct supervisor. Reporting protects patients and keeps the workplace accountable.

#483 · Professional Practice

What does lifelong learning mean for an MLA?

Answer

Lifelong learning means continuing to build knowledge and skills throughout a career, not stopping after certification. This can include attending workshops, reading updated guidelines, or learning new equipment. It helps MLAs keep up with changes in lab technology and best practices.

#484 · Professional Practice

Why does CSMLS require continuing competency activities for certified members?

Answer

Continuing competency activities help ensure that certified professionals stay current with new procedures, safety standards, and technology. It shows a commitment to quality patient care over time. Without ongoing learning, skills and knowledge can become outdated.

#485 · Professional Practice

What is evidence-informed practice?

Answer

Evidence-informed practice means basing decisions and procedures on the best available research and proven guidelines, not just habit or personal opinion. For an MLA, this could mean following updated collection procedures based on new studies. It helps ensure care is safe and effective.

#486 · Professional Practice

Give an example of evidence-informed practice in specimen collection.

Answer

Following an updated policy on tourniquet time during blood draws because research shows prolonged application can affect test results. Updating hand hygiene steps based on new infection control studies is another example. Both show practice adjusting based on evidence, not tradition alone.

#487 · Professional Practice

What is a good first step in solving a workplace problem, like a broken piece of equipment?

Answer

The first step is usually to identify the exact problem and check if there is a simple, approved fix within your scope. If not, report it to the appropriate person, such as a supervisor or lab technologist. Trying to fix technical equipment outside your training can create bigger problems.

#488 · Professional Practice

What should an MLA do if they are unsure whether a task is within their scope of practice?

Answer

They should stop and ask a supervisor or senior team member before performing the task. Guessing or assuming can put patients at risk and create legal issues for the worker. When in doubt, it is always safer to check first.

#489 · Professional Practice

Why is honesty considered a core professional value in lab work?

Answer

Honesty ensures that results, records, and communications can be trusted by patients and healthcare teams. If an MLA hides a mistake, like a mislabeled sample, it could lead to a wrong diagnosis or treatment. Trust in the healthcare system depends on workers being truthful, even when it is uncomfortable.

#490 · Professional Practice

What is meant by respecting patient autonomy?

Answer

Respecting patient autonomy means recognizing a patient's right to make their own choices about their care, including refusing a procedure. An MLA should explain the collection process clearly but not pressure a patient who refuses. Their decision should be documented and reported to the appropriate person.

#491 · Professional Practice

A patient refuses a blood draw. What is the professional response?

Answer

The MLA should respect the patient's decision without arguing or pressuring them. They should calmly explain why the test was ordered, but ultimately accept the refusal. The refusal must then be documented and reported to the supervising nurse or physician.

#492 · Professional Practice

What is the purpose of a professional regulatory body like CSMLS?

Answer

A regulatory or certifying body sets standards for education, ethics, and competency in a profession. It helps protect the public by ensuring certified workers meet consistent expectations. It also provides a process for handling complaints or violations of professional standards.

#493 · Professional Practice

What should an MLA do if asked by a supervisor to perform a task outside their training?

Answer

The MLA should respectfully explain that the task is outside their scope of practice or training. They can ask for guidance on who is qualified to perform it instead. Performing the task anyway could put the patient and the MLA at risk.

#494 · Professional Practice

Why is punctuality considered part of professional behaviour?

Answer

Being on time shows respect for coworkers, patients, and the smooth operation of the lab. Late arrivals can delay specimen collection schedules and put extra pressure on the team. Reliability builds trust with both patients and colleagues.

#495 · Professional Practice

What is meant by maintaining objectivity in the lab?

Answer

Objectivity means making decisions based on facts and procedures rather than personal feelings or bias. For example, treating every patient the same way regardless of personal opinions about them. This ensures fair and consistent care for all patients.

#496 · Professional Practice

How should an MLA handle a personal disagreement with a coworker during a shift?

Answer

The MLA should stay professional and keep the disagreement from affecting patient care or teamwork. It is best to address the issue calmly in private, away from patients, or bring it to a supervisor if needed. Personal conflicts should never be discussed in front of patients.

#497 · Professional Practice

What is the risk of accepting gifts from patients?

Answer

Accepting gifts can create a conflict of interest or the appearance of favoritism toward certain patients. Many workplaces have policies limiting or banning gift acceptance for this reason. Politely declining protects both the worker's professionalism and the fairness of patient care.

#498 · Professional Practice

What does it mean to practice cultural safety in the lab?

Answer

Cultural safety means being aware of and respectful toward a patient's cultural background, beliefs, and needs during care. This could include understanding why a patient might be hesitant about a procedure due to cultural reasons. It helps build trust and reduces barriers to care for diverse patients.

#499 · Professional Practice

Why should an MLA avoid gossiping about patients or coworkers, even outside of work?

Answer

Gossip can spread private information and damage trust, even if it feels harmless. It can also lead to confidentiality breaches if patient details are shared outside of work. Professional behaviour extends beyond the workplace to protect patient privacy and workplace relationships.

#500 · Professional Practice

What is a simple problem-solving approach an MLA can use when facing an unclear situation at work?

Answer

A good approach is to identify the problem clearly, think of possible safe solutions within scope, and ask for help if unsure. Following established procedures and reporting lines usually leads to the safest outcome. Rushing to solve something outside training can create bigger risks for patients.